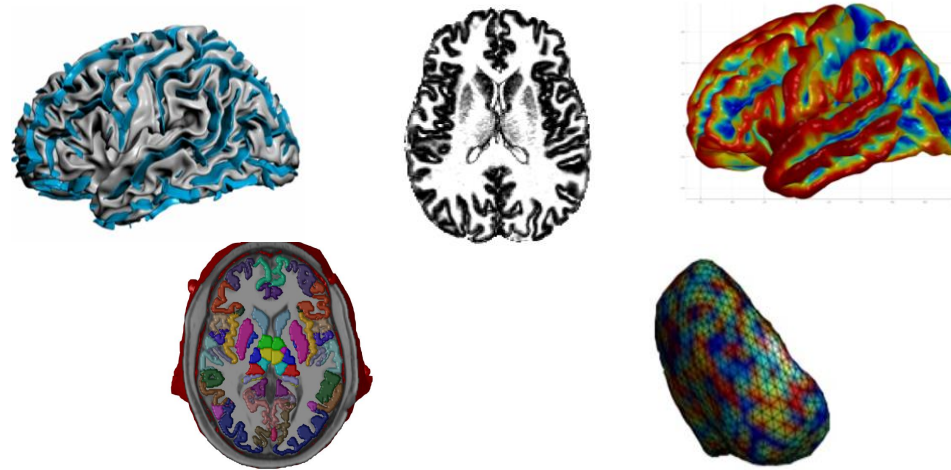




MRI Processing Theory: Structural Morphometry

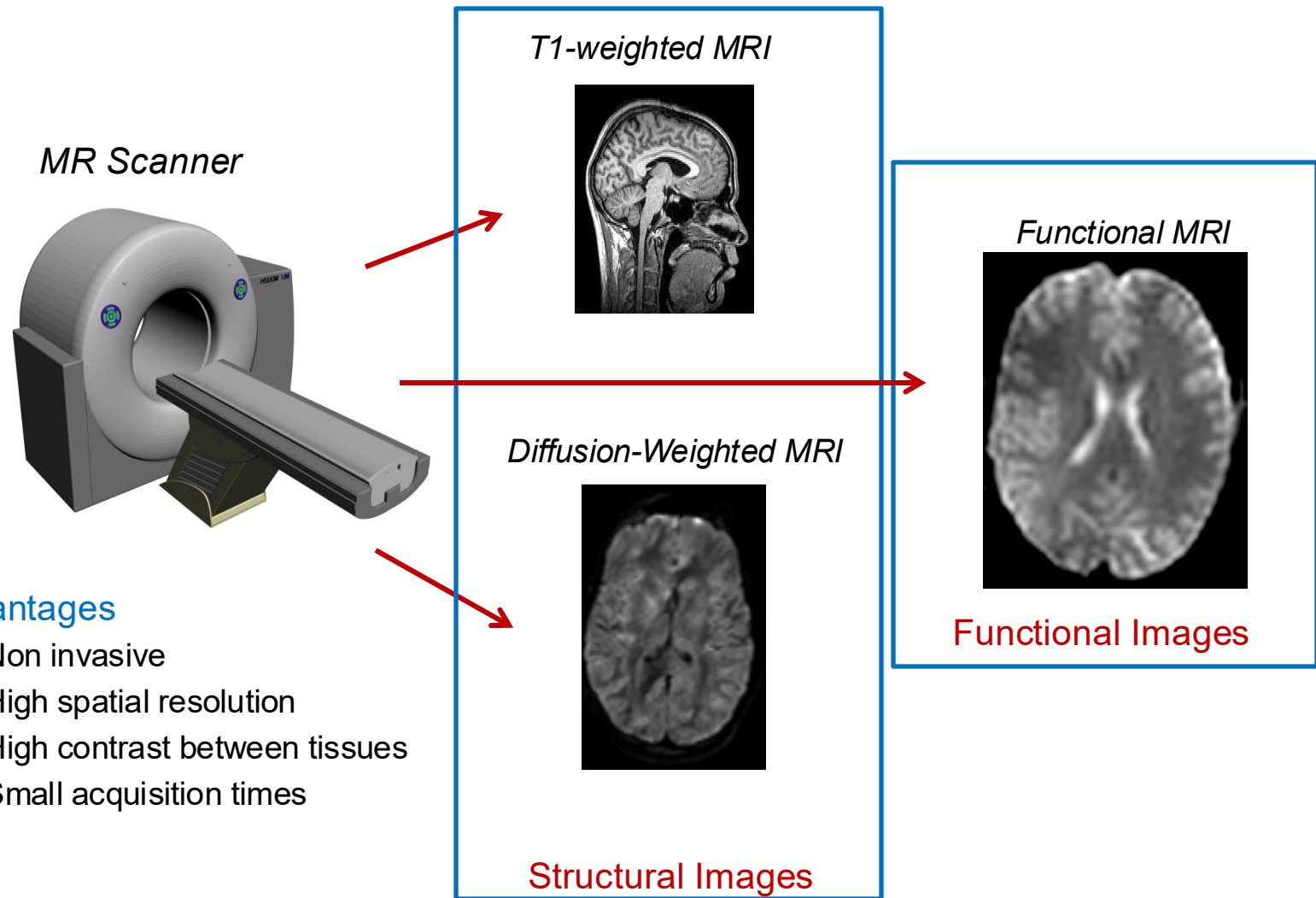
Yasser Alemán-Gómez

Radiology department. CHUV, Switzerland
October 1st, 2026

Outline

- Introduction to MRI Processing
- Region of Interest (ROI) Analysis
- Voxel-Based Morphometry (VBM)
- Surface Based Morphometry (SBM)

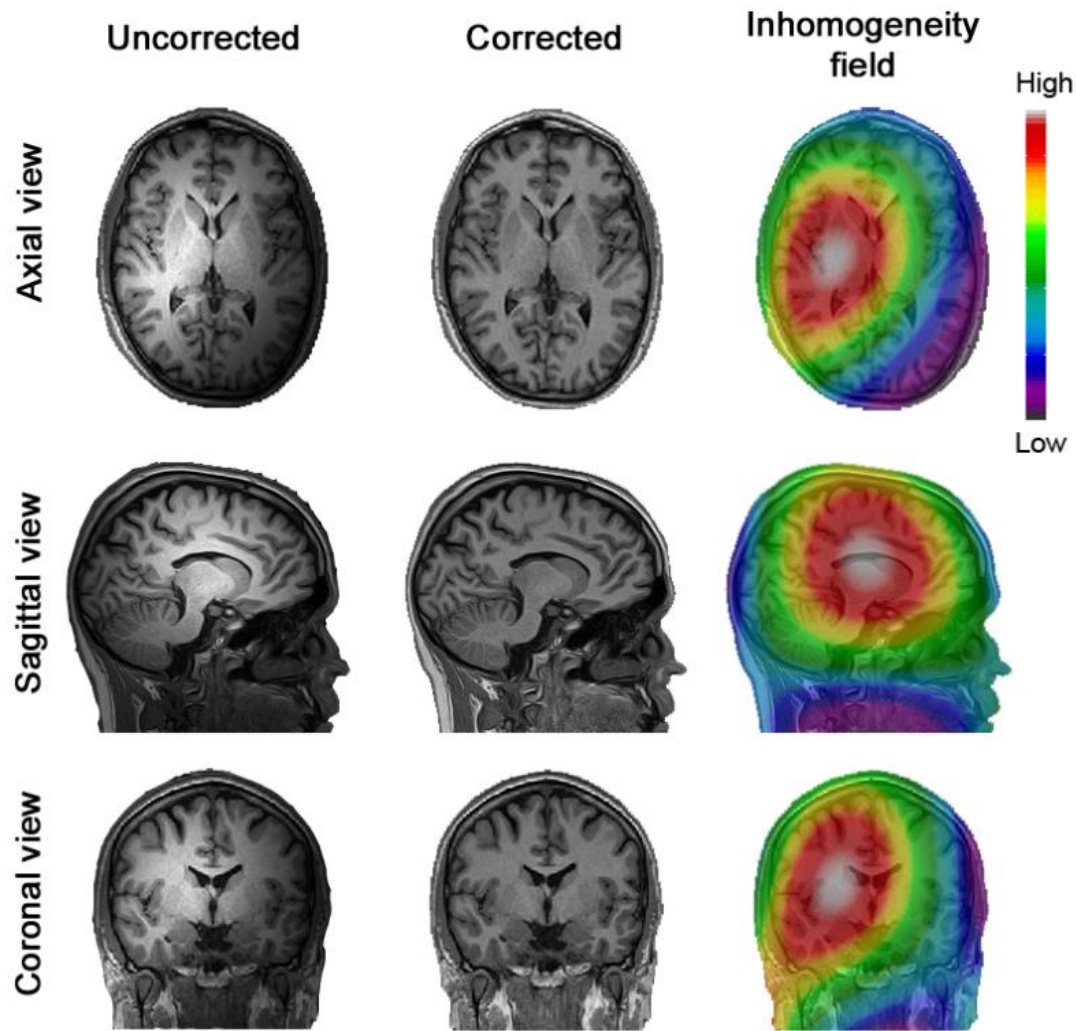
Introduction: MRI processing



■ Advantages

- ❑ Non invasive
- ❑ High spatial resolution
- ❑ High contrast between tissues
- ❑ Small acquisition times

Bias field correction



ANTS
(*N4BiasFieldCorrection*, *C++*)

Region of Interest (ROI) Analysis

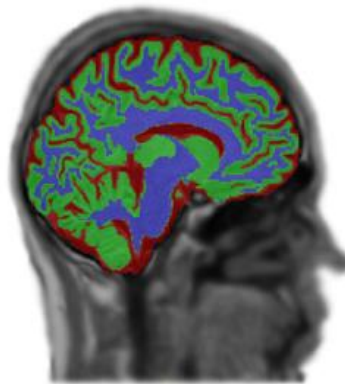
T1-Weighted image



Tissue
Segmentation



Brain tissues



Available methods

FSL

(fast, C++)

ANTs

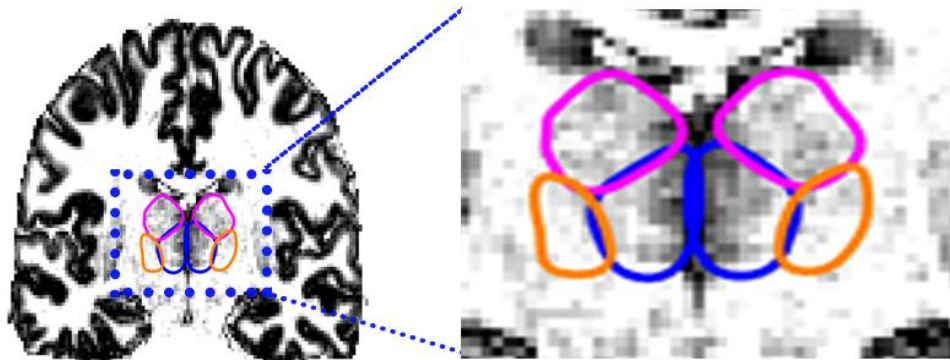
(Atropos, C++)

SPM

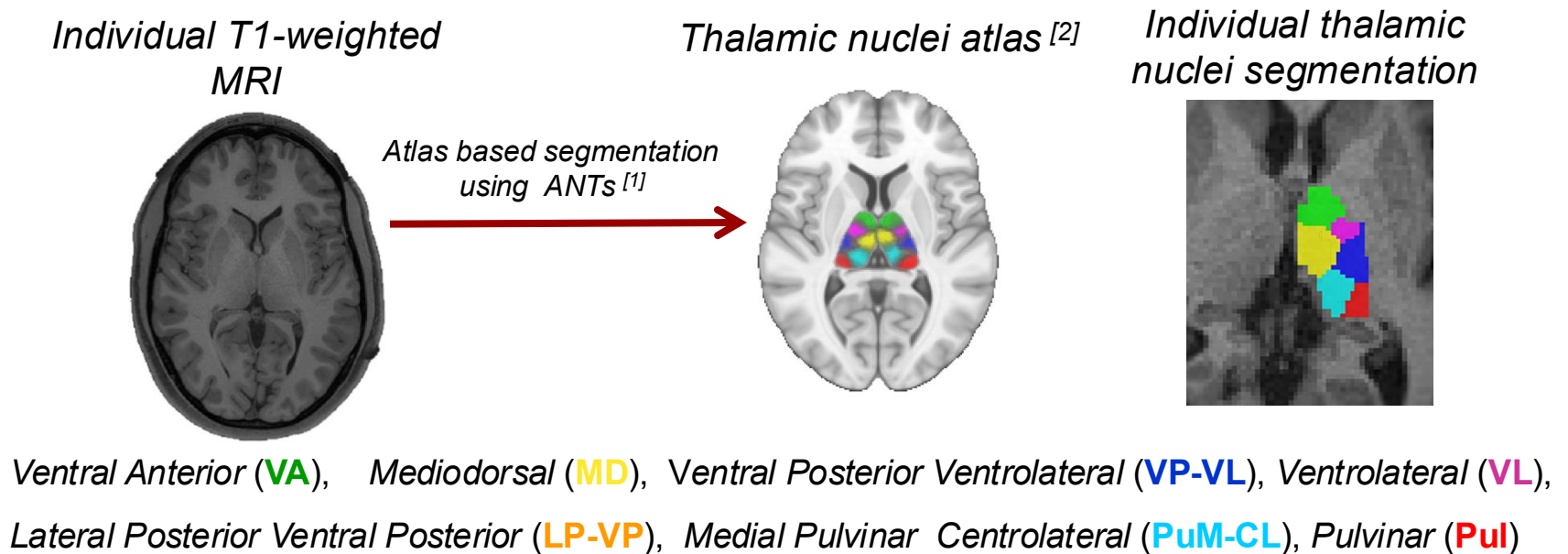
(spm_segment, Matlab)

NiSeg

(partial_volume_estimation, Python)



Region of Interest (ROI) Analysis



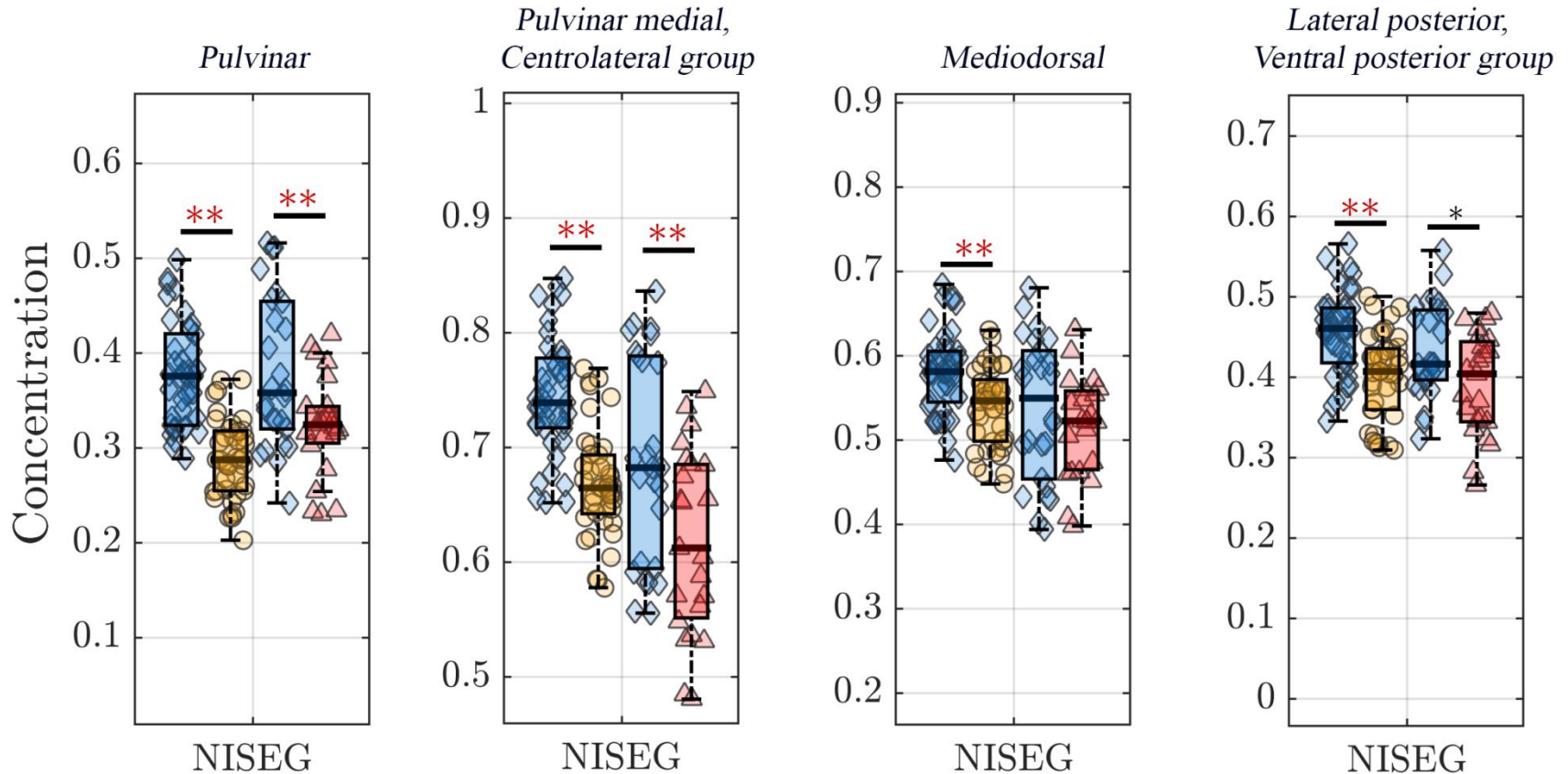
■ Advantages

- It works in subject space

■ Drawbacks

- Previously defined ROI

Group differences in gray matter for the thalamic nuclei



◆ Healthy controls

● Patients with early psychosis

▲ Patients with schizophrenia

* Significant differences between groups before FDR correction

** Differences that remained significant after the FDR correction

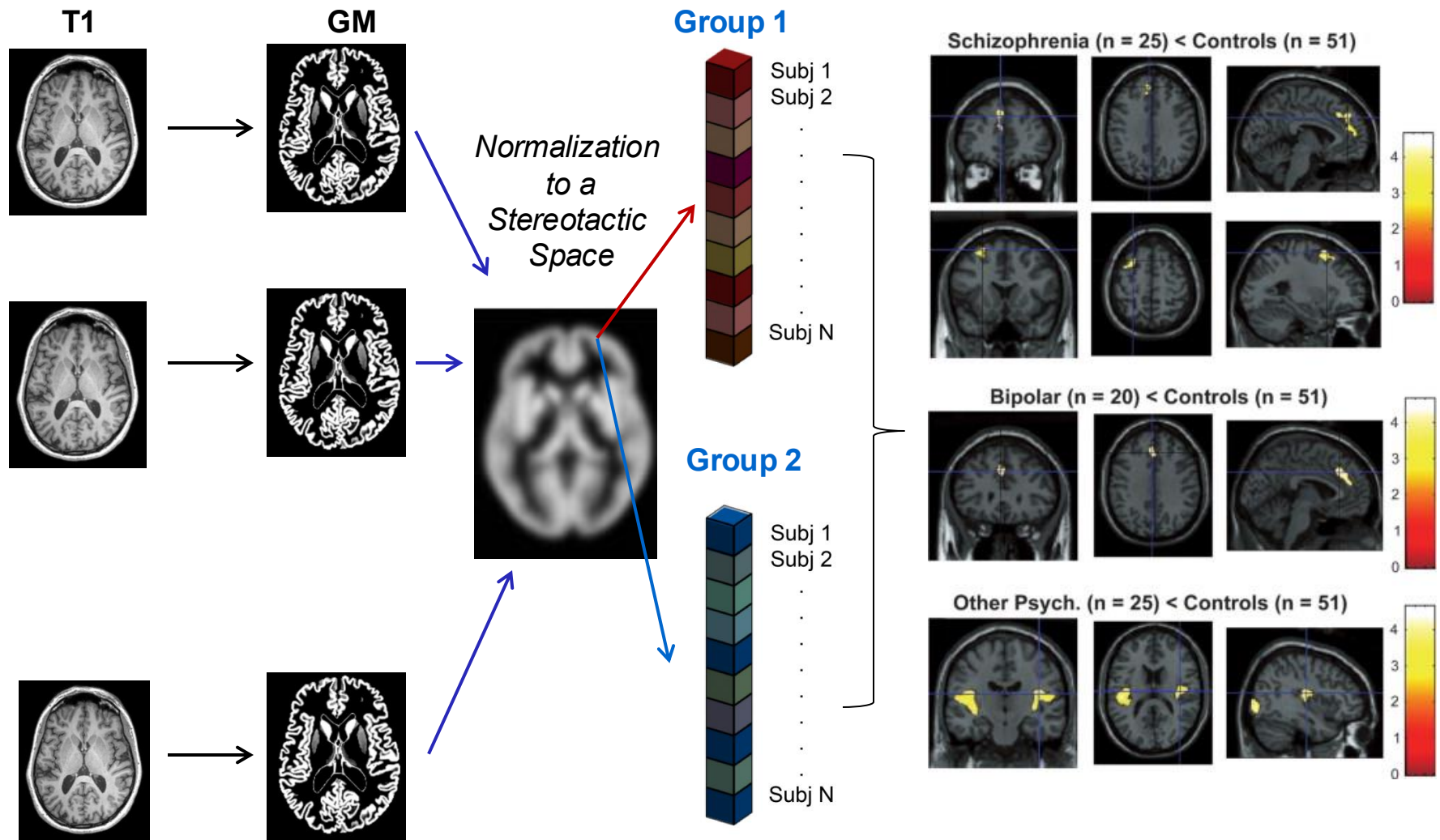
Alemán-Gómez Y, Human Brain Mapping, 2020



A 3D model of a human head, showing the brain in blue and the skull in red. The brain is positioned within the skull, and the skull is shown in a cross-section view.



Voxel-Based Morphometry (VBM)



Voxel-Based Morphometry (VBM)

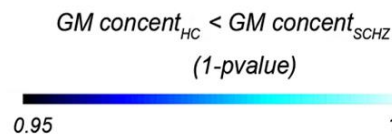
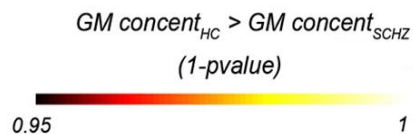
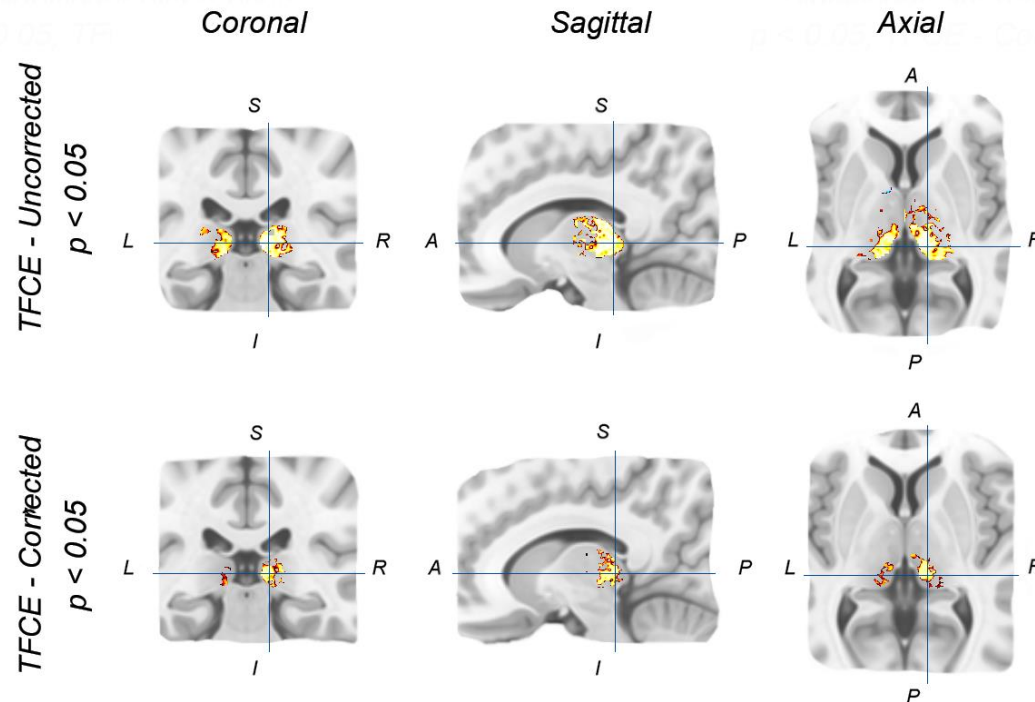
Significant differences in GM concentration between groups

Advantages

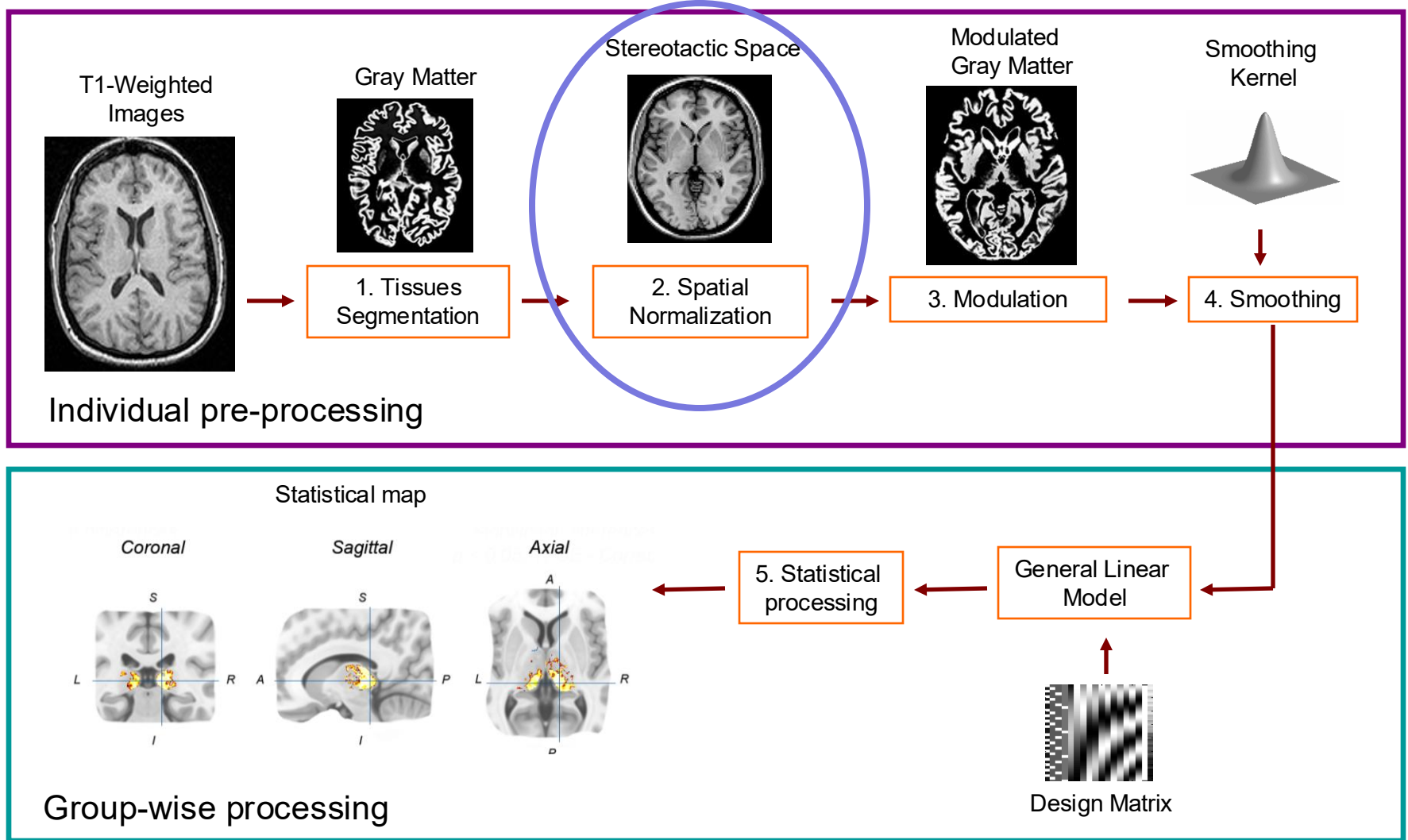
- No apriori spatial information is needed

Drawbacks

- Strong alignment between subjects in stereotactic space

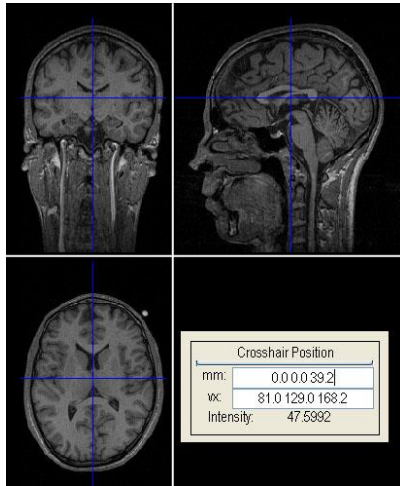


Voxel-based morphometry

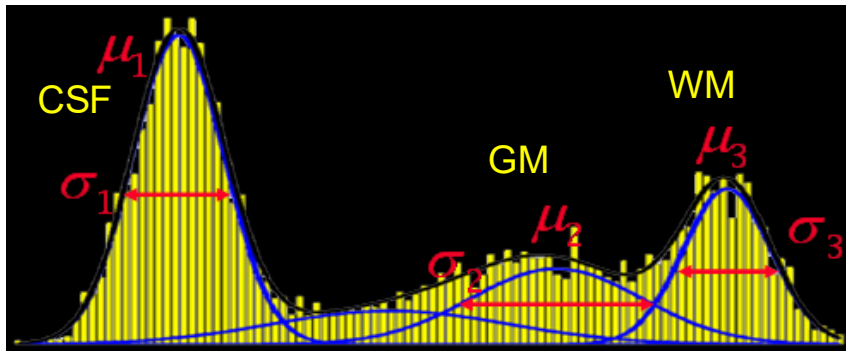


GM concentration vs GM probability

T1-weighted image

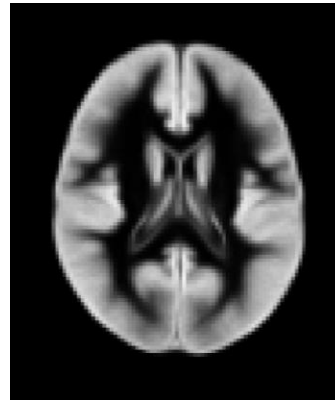


Gaussians Mixture Model

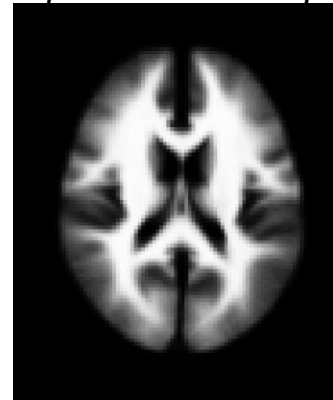


A priori probabilistic maps

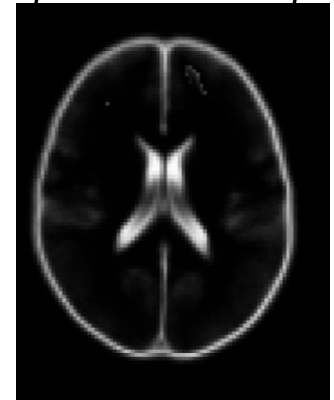
Gray Matter probabilistic map



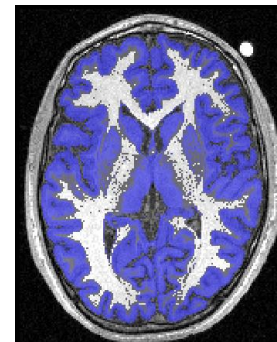
White Matter probabilistic map



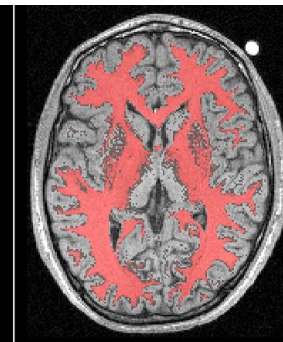
Cerebrospinal fluid probabilistic map



Gray Matter



White Matter

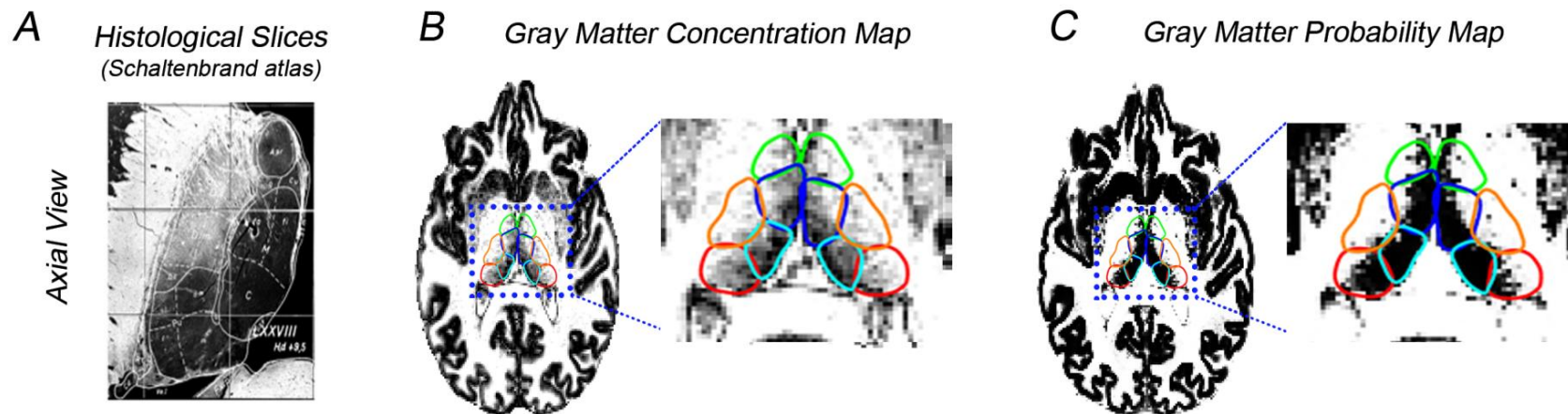


Cerebrospinal fluid



GM concentration vs GM probability

- Thalamus is a mixture of gray and white matter
- Gray matter assessment in the thalamus has been done by using probability-based approaches
- Partial Volume modeling

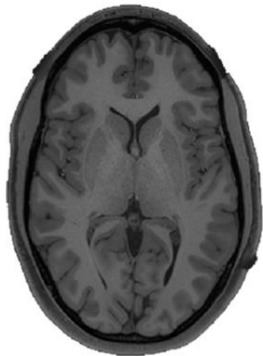


$$I_{VOXEL} = C_{GM} \cdot I_{GM} + C_{WM} \cdot I_{WM} + C_{CSF} \cdot I_{CSF}, \quad C_{GM} + C_{WM} + C_{CSF} = 1$$

Gray Matter (GM), White Matter (WM) and CerebroSpinal Fluid(CSF) concentrations

Spatial Normalization

T1-weighted image

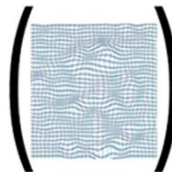


T1-weighted template

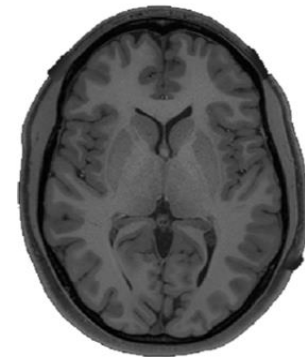


*Spatial
Registration*

*Spatial transformation
matrix*

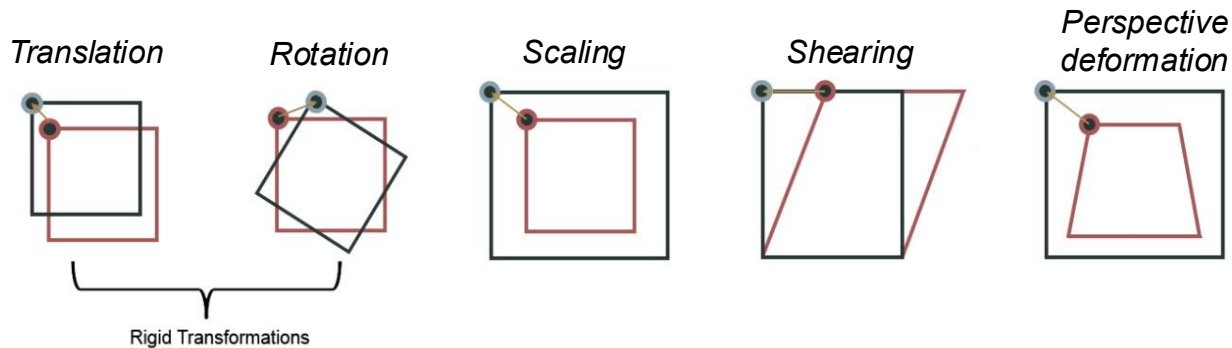


*Spatially normalized
T1-weighted image*



Spatial Normalization

1. Affine transformation: Matches overall position and size

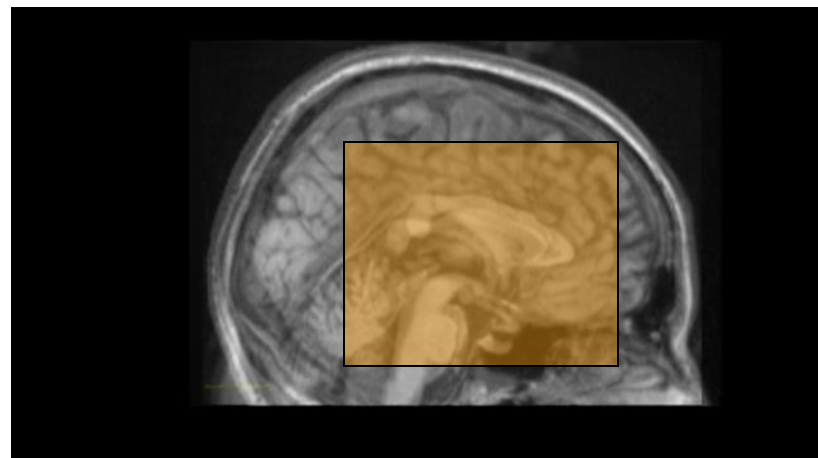
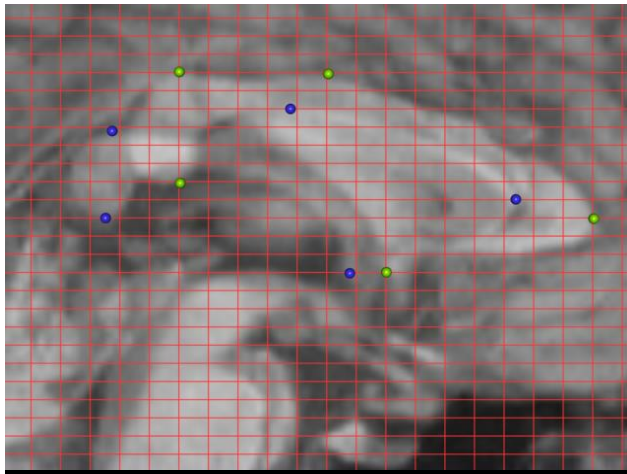


FSL
(*flirt*, C++)

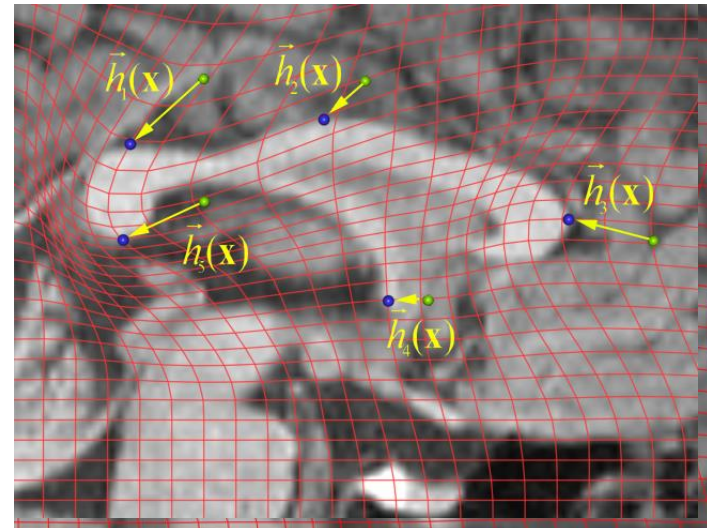
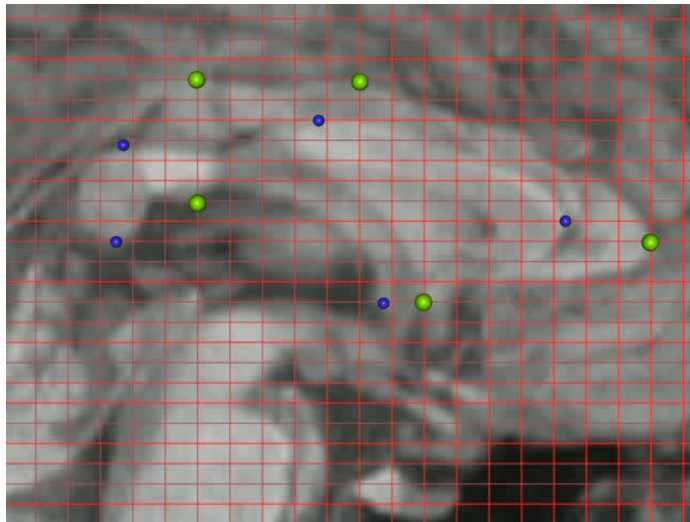
ANTs
(*antsRegistration*, C++)

SPM
(*spm_coreg*, Matlab)

Affine registration between 2 images



Spatial Normalization (Non-linear registration)



Cost function:

$$E = \gamma C[I_1, I_2(h)] - \alpha \text{Deformación}$$

Similarity between
images

Apriori information
about the
deformation field

Available methods [1]

FSL
(*fnirt*, C++)

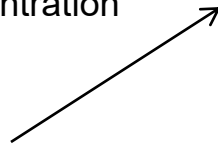
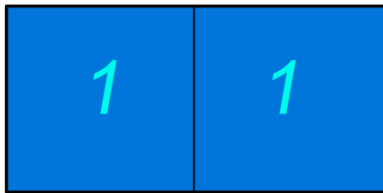
ANTs
(*antsRegistration*, C++)

Modulation

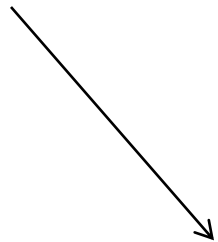
Preserve probability/concentration by multiplying the warped (normalised) tissue intensities by the Jacobian of the spatial transformation

Native Space:

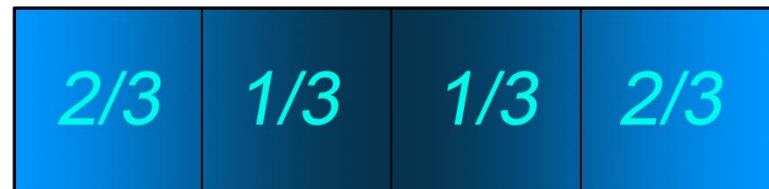
Intensity = tissue probability/concentration



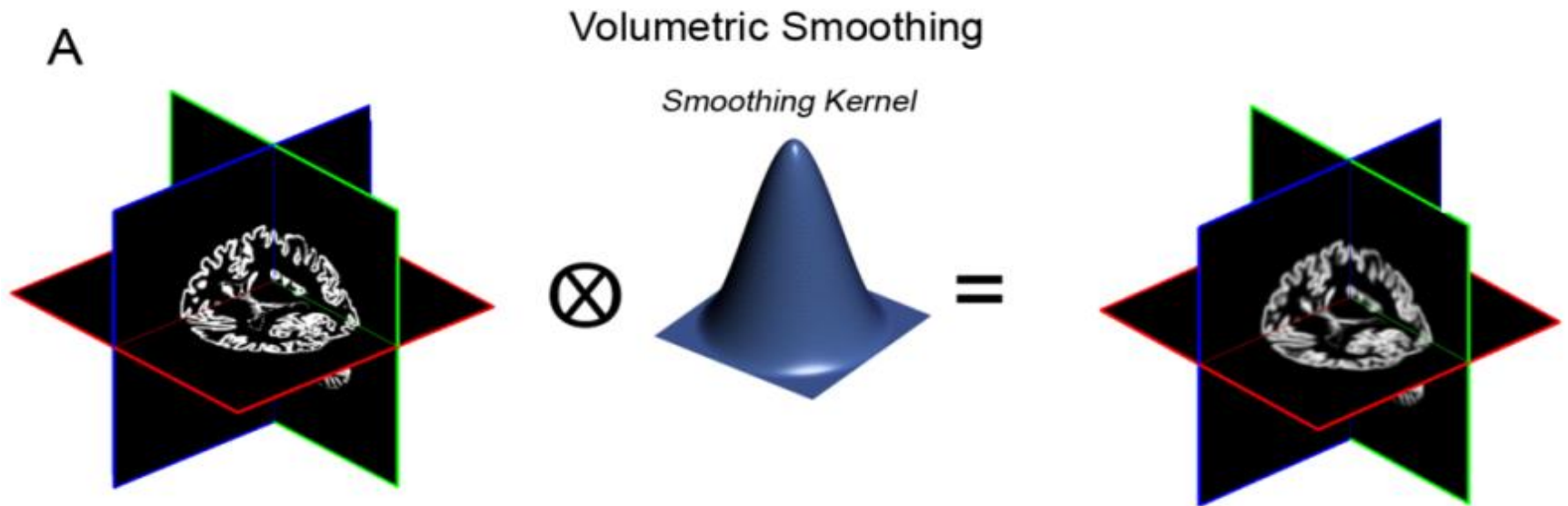
Unmodulated



Modulated



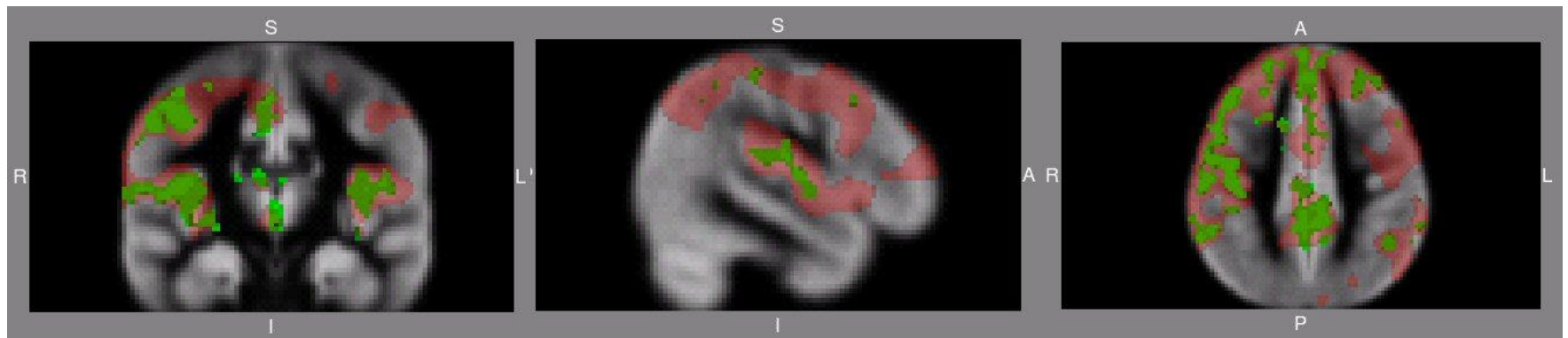
Volumetric smoothing



Smoothing

- The data will be more Gaussian and closer to a continuous random field for larger kernels
- Results will be rough and noise-like if too little smoothing is used
- Too much will lead to distributed, indistinct blobs
 - Kernel size recommended: $\geq 6\text{mm}$ and $\leq 12\text{mm}$

Results for different smoothing Kernels



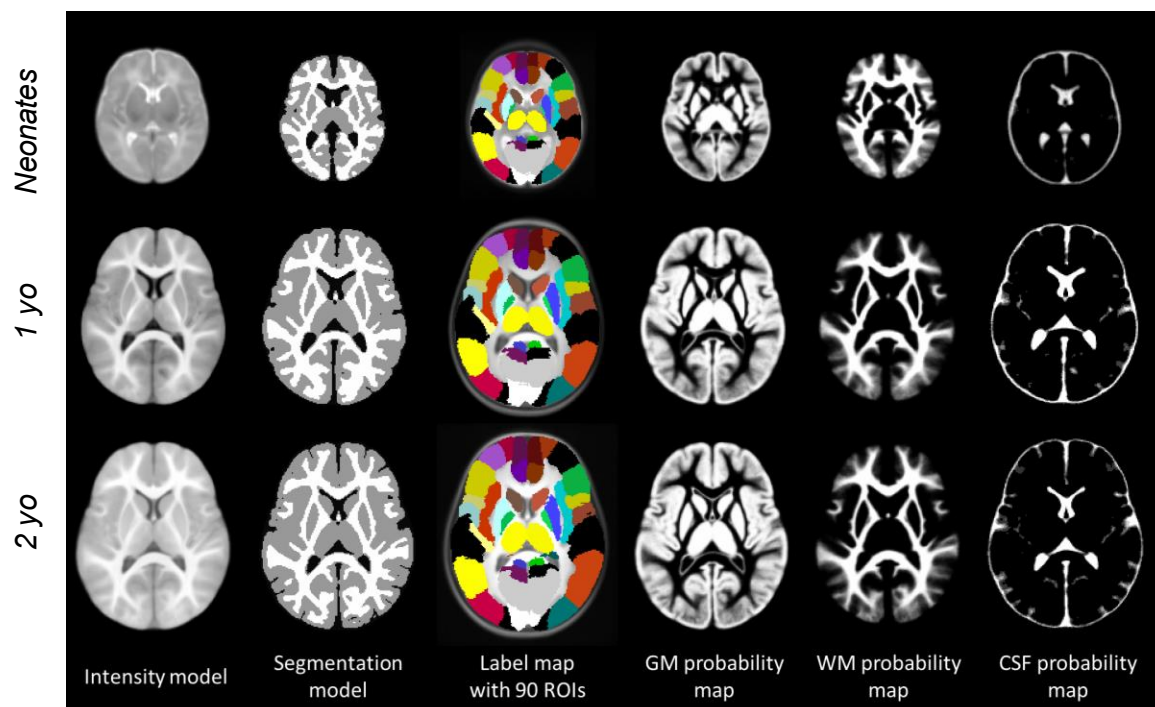
Green: FWHM = 8mm

Red: FWHM = 12mm

Population-specific template

- Custom template building

- Tissues contrast
- Age range
- Gender
- Ethnic



- Symmetric group-wise diffeomorphic normalization (SyN) ^[1] and FNIRT

ANTs

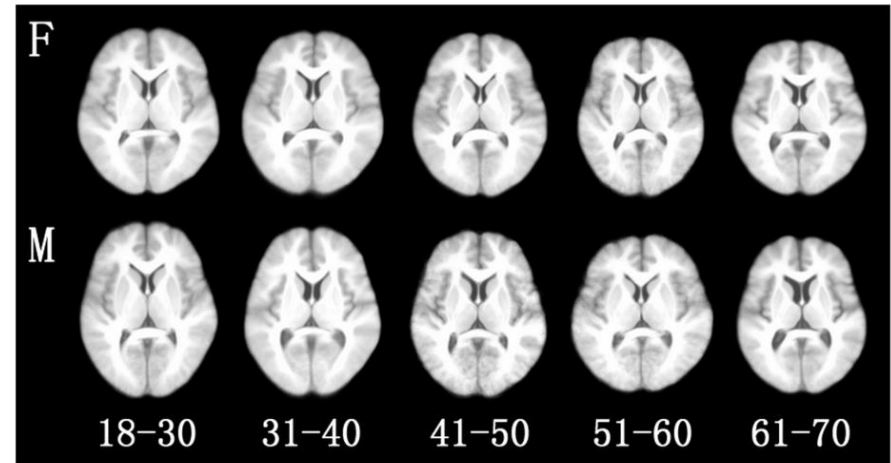
(*buildtemplateparallel.sh*, C++)

FSL

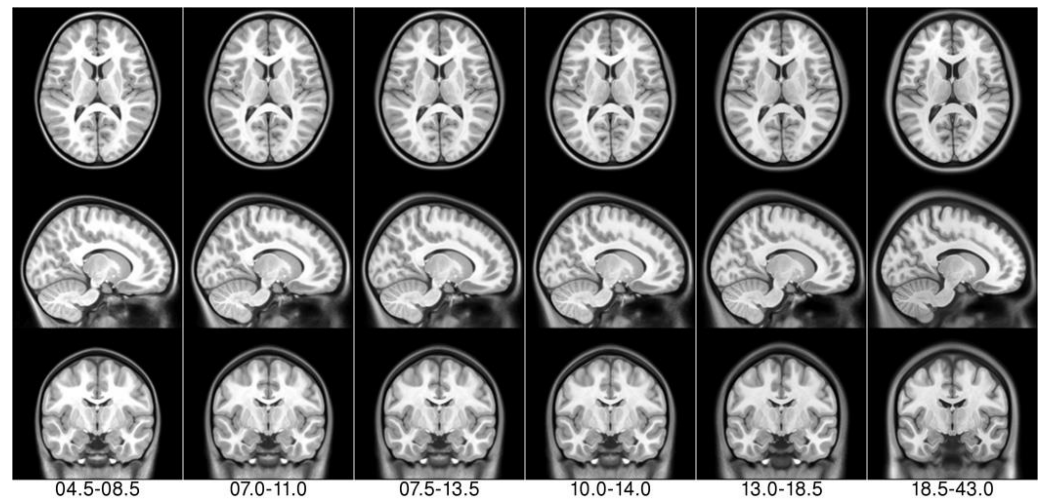
(*fslvbm_2_template*, C++)

Population-specific template

*Probabilistic MRI Brain Anatomical Atlases
Based on 1000 Chinese Subjects [1]*



*Unbiased Average Age-Appropriate
Atlases for Pediatric Studies [2]*



[1] Xing et al 2013. Plos One

[2] Fonov et al 2011. Neuroimage

Interpretation of VBM results

Reference



Mis-classification



Mis-registration



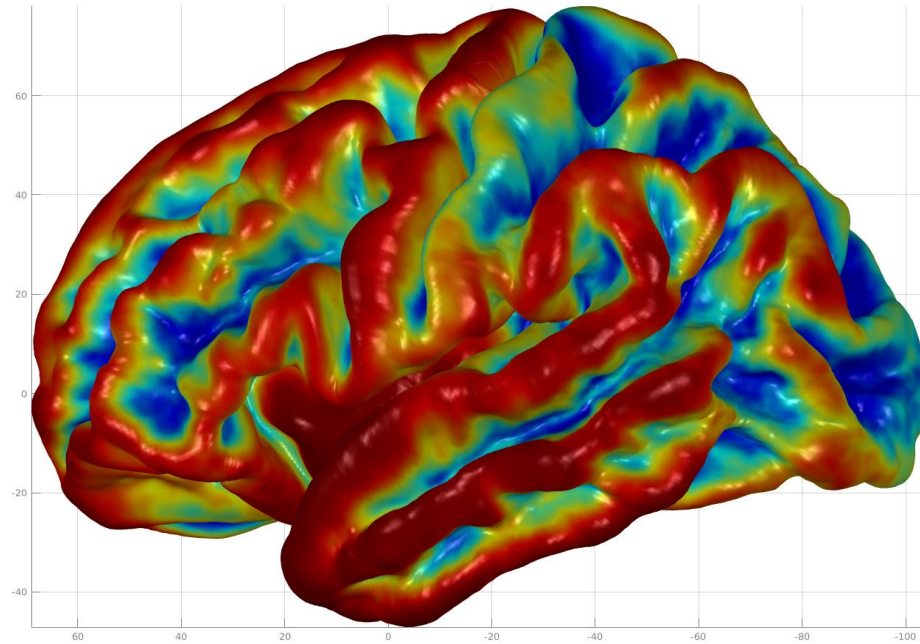
Thickening or thinning



Unfolding



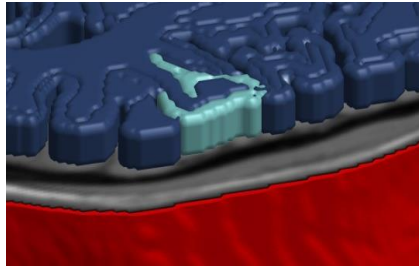
- In essence VBM is Statistical Parametric Mapping of regional segmented tissue density or volume
- The exact interpretation of gray matter density or volume is complicated, and depends on the preprocessing steps used
 - It is not interpretable as neuronal packing density or other cytoarchitectonic tissue properties
 - The hope is that changes in these microscopic properties may lead to macro- or mesoscopic VBM-detectable differences



Surface Based Morphometry (SBM)

Volume as a composite measure

Cortical region



Box



Pyramid

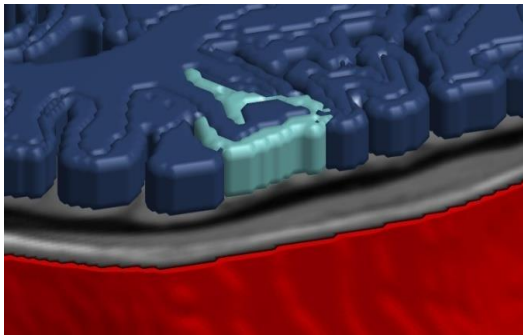


Teapot

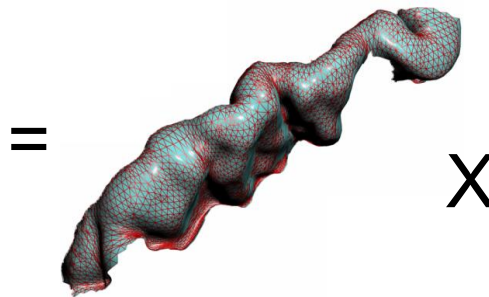


Same volume different shape

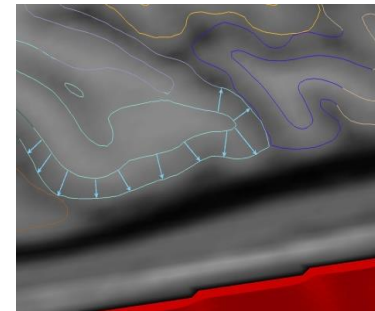
Volume



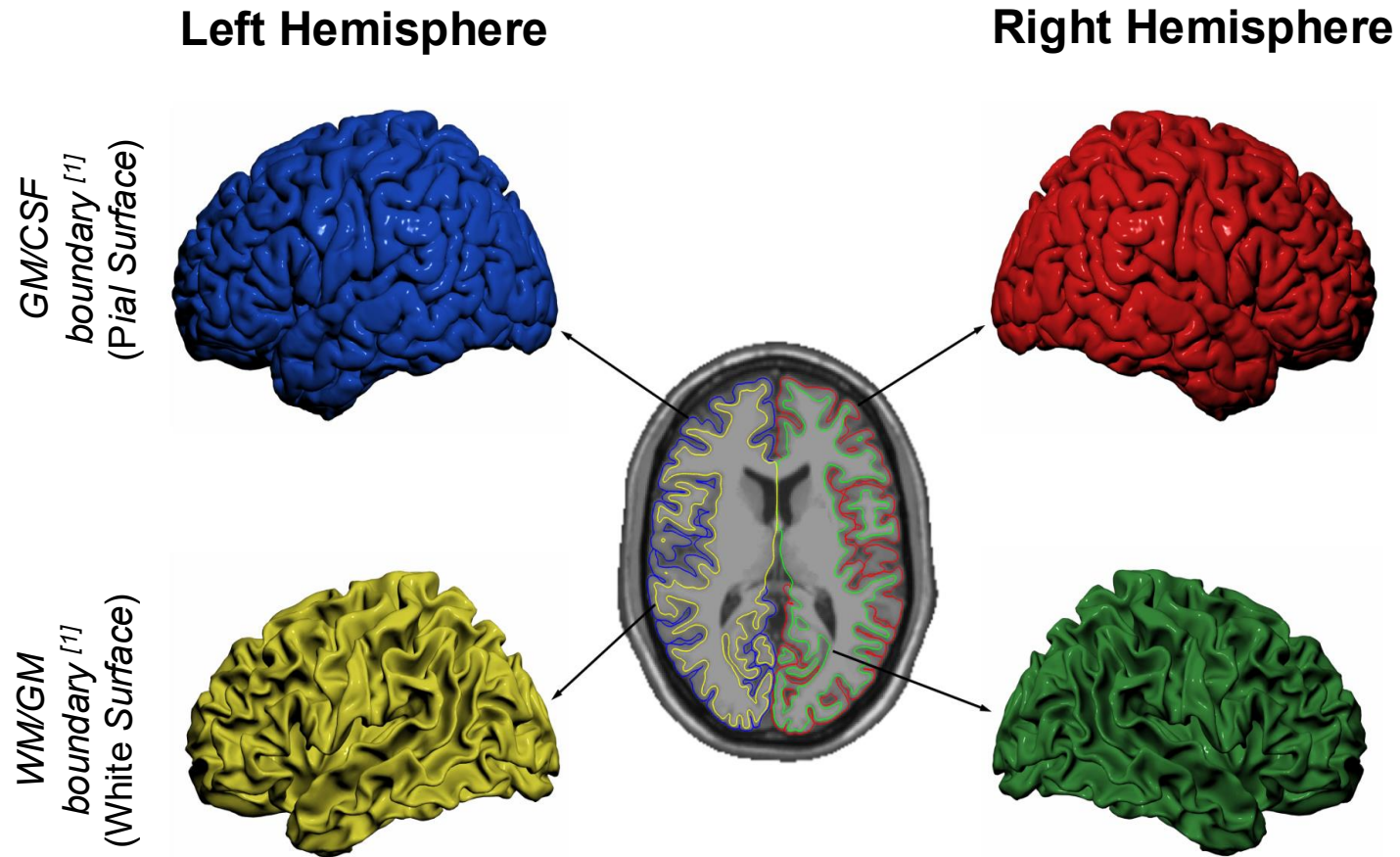
Surface Area



Cortical thickness



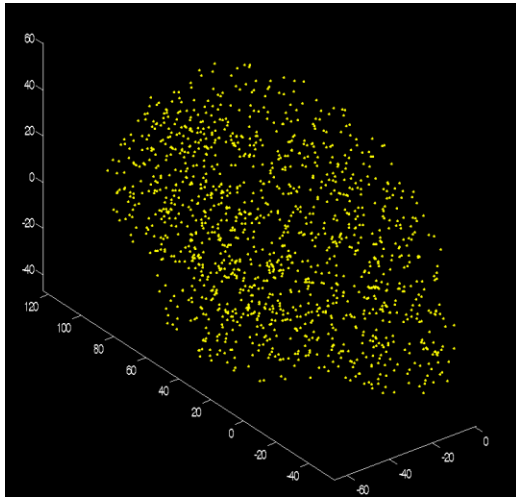
Cortical surfaces



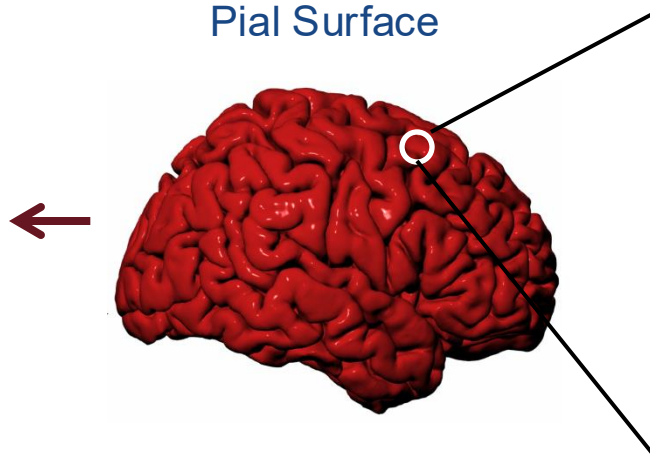
[1] Davatzikos et al. 1996. *IEEE Transactions on Medical Imaging*

Basic concepts:

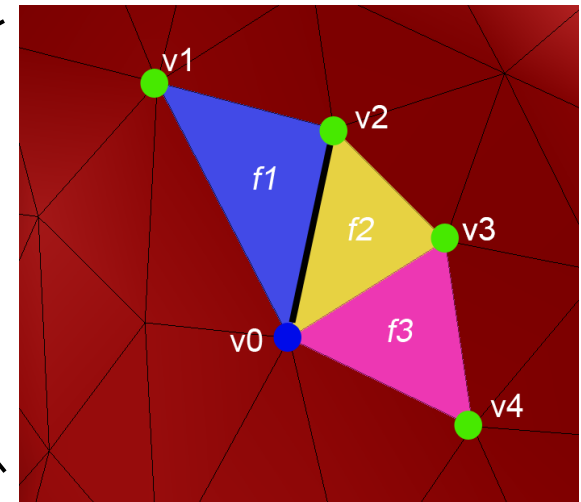
Pial Surface vertices



Pial Surface



Faces



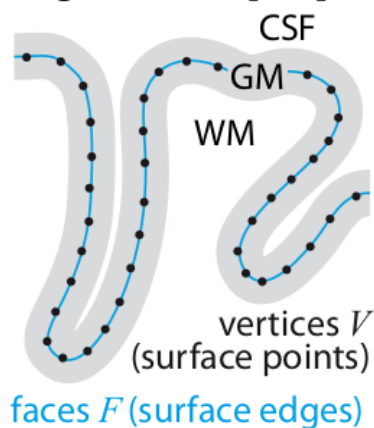
$$f1 = (v0, v1, v2)$$

$$f2 = (v0, v2, v3)$$

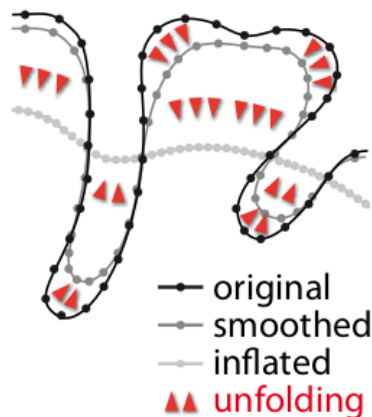
$$f3 = (v0, v3, v4)$$

Operations with surfaces

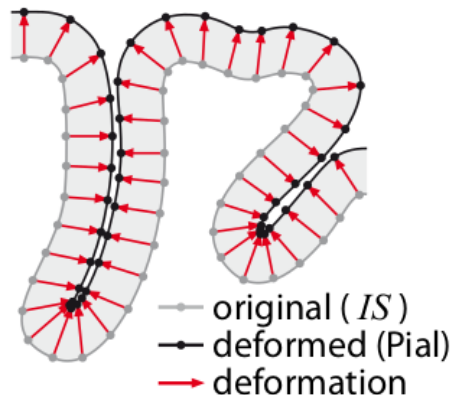
original $CS = [V, F]$



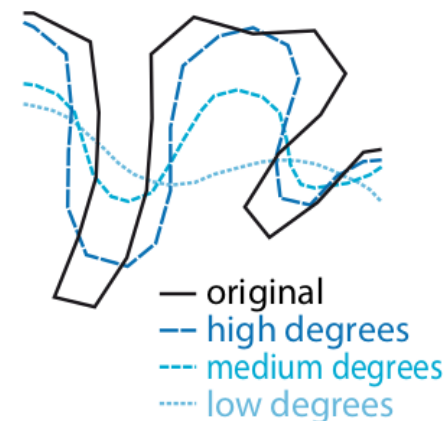
a) Smoothing & Inflation



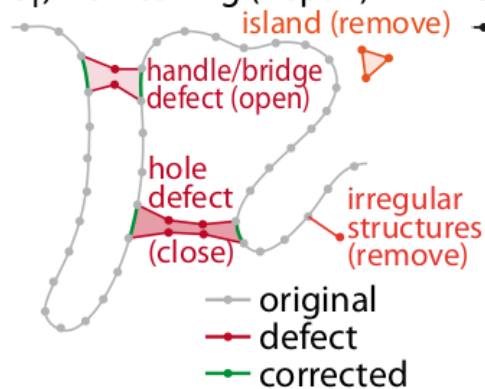
b) Deformation



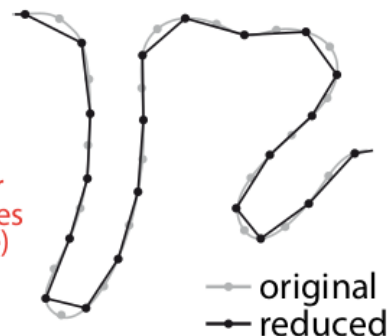
d) Parameterization



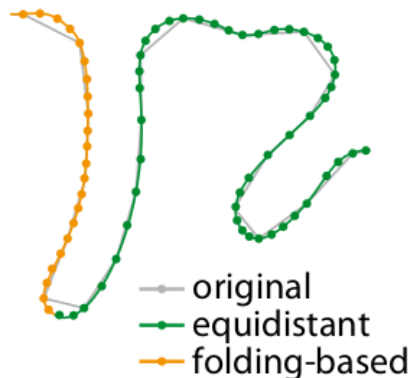
c₁) Remeshing (Repair)



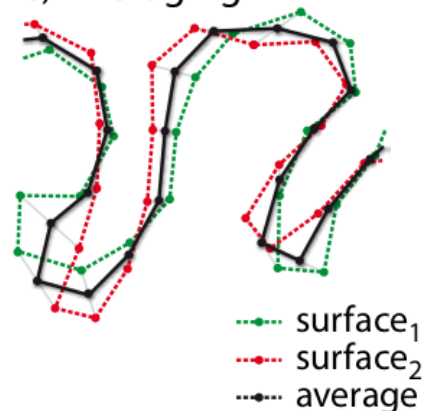
c₂) Remeshing (Reduction)



c₃) Remeshing (Refinement)

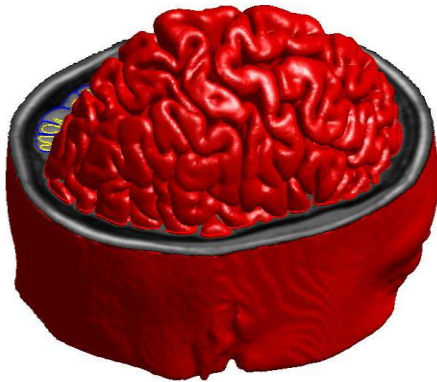


e) Averaging

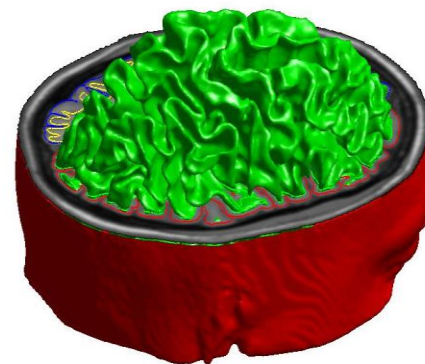


Cortical surfaces

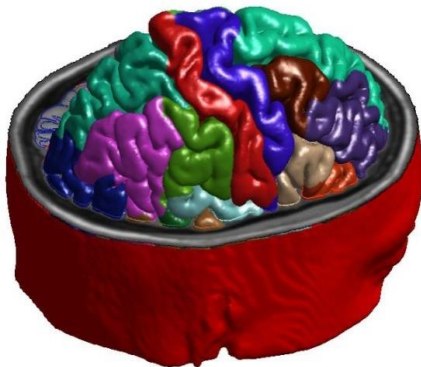
**GM/CSF
(Pial Surface)**



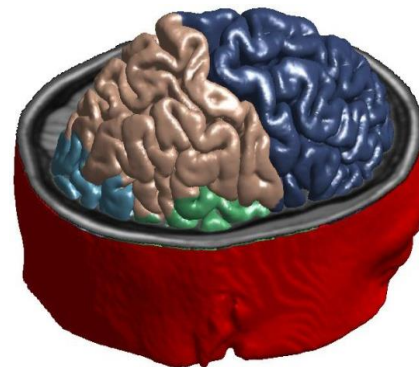
**WM/GM
(White Surface)**



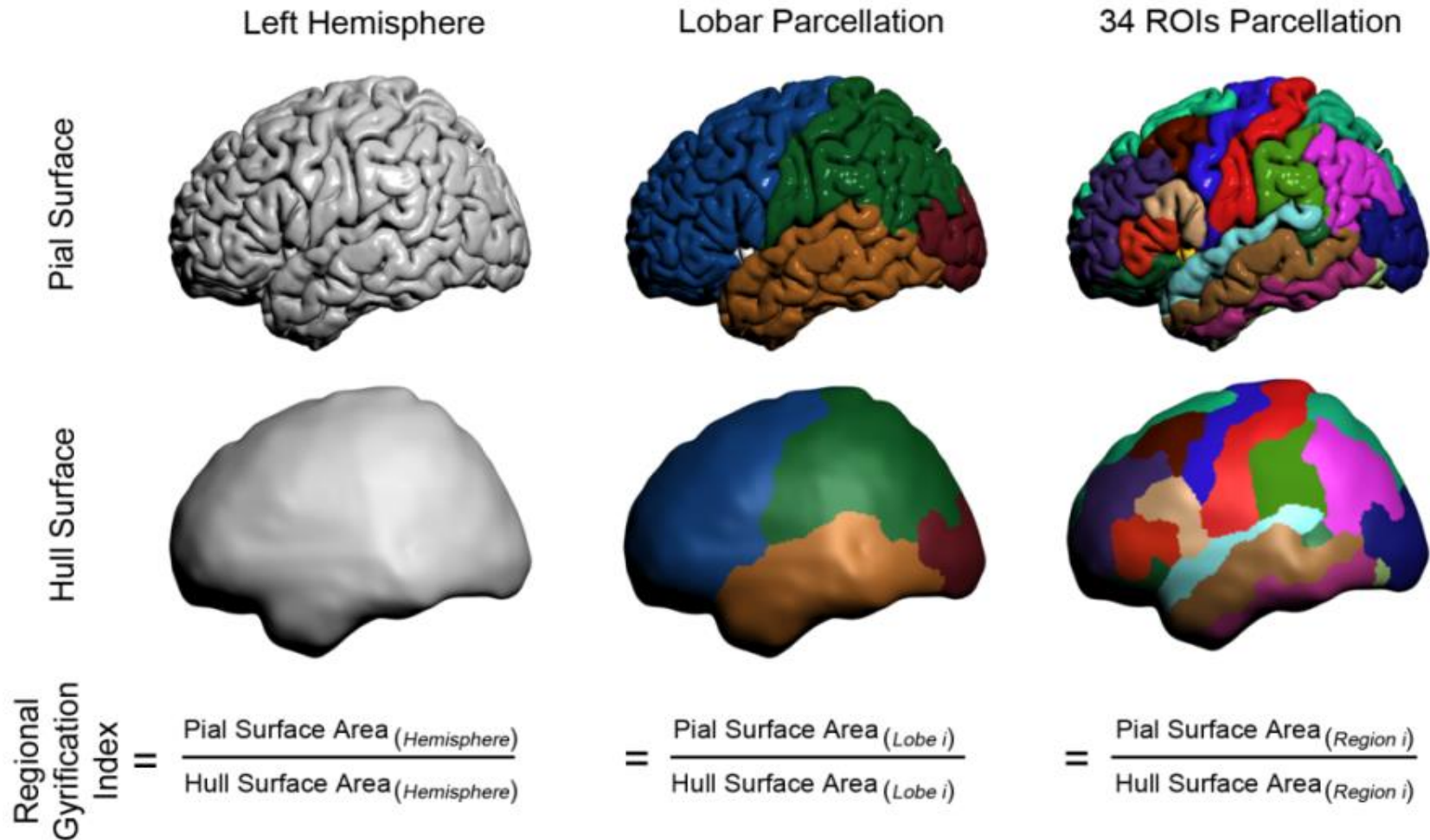
**Cortical parcellation
(Desikan et al 2006)**



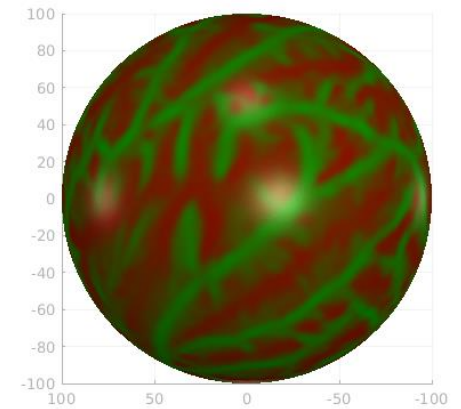
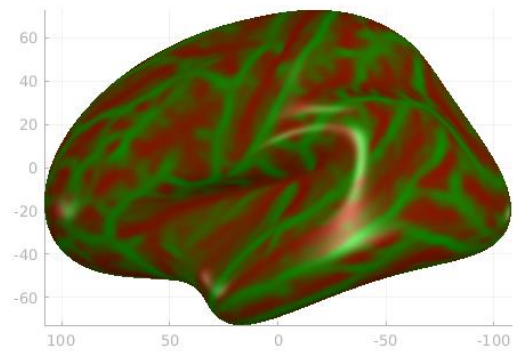
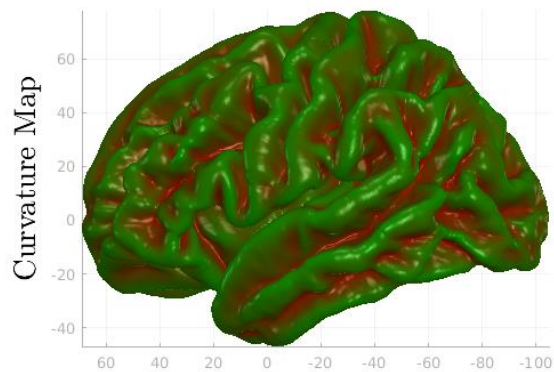
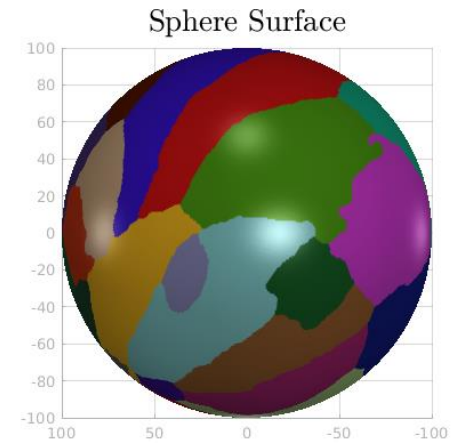
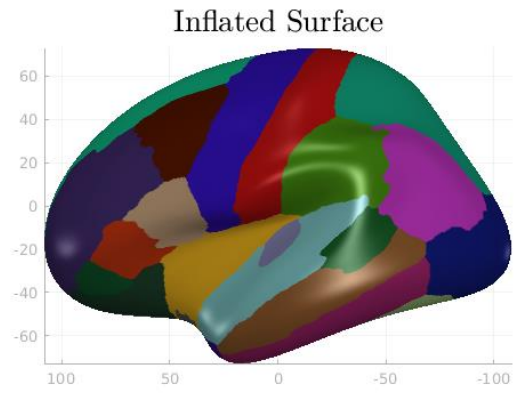
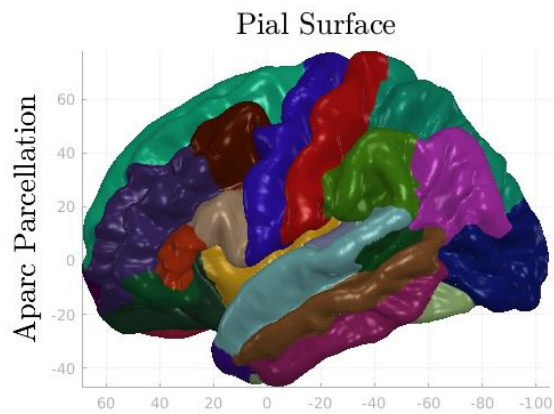
**Lobar
parcellation**



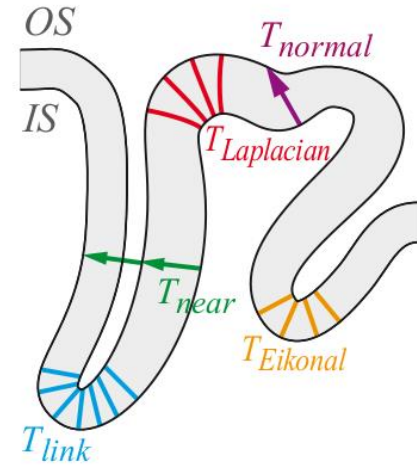
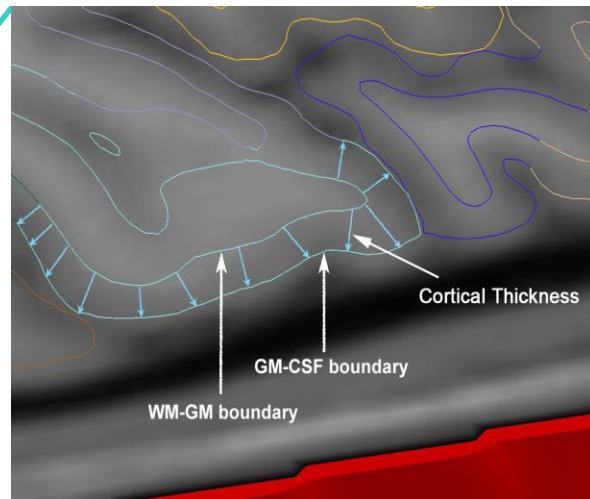
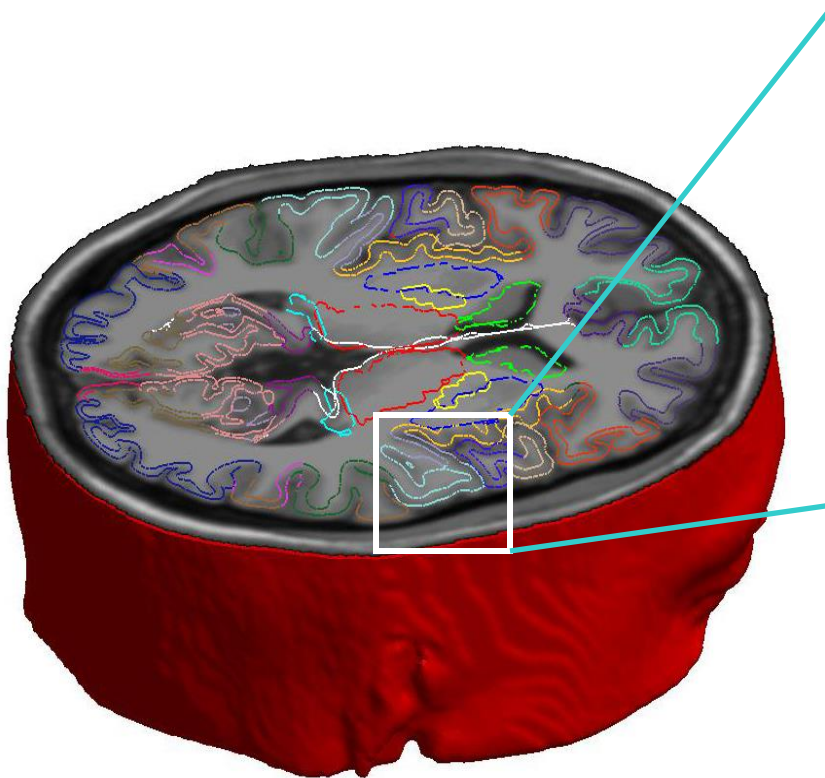
Regional metrics



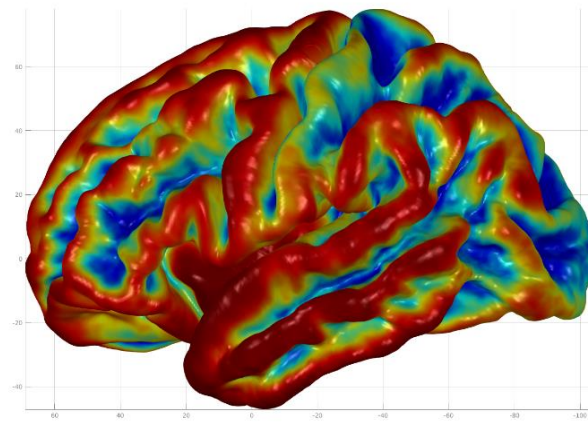
Useful surfaces



Cortical thickness

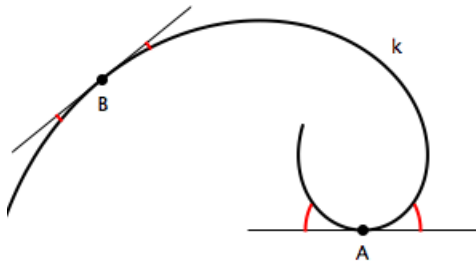


Cortical thickness map

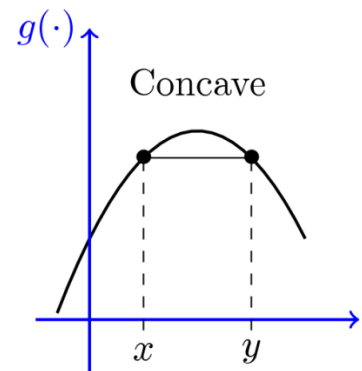
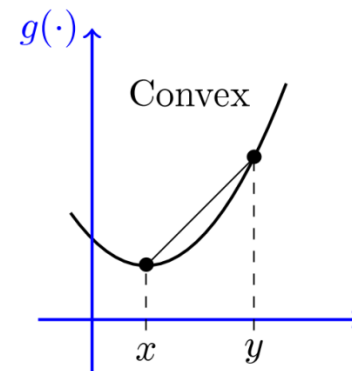


Curvature maps

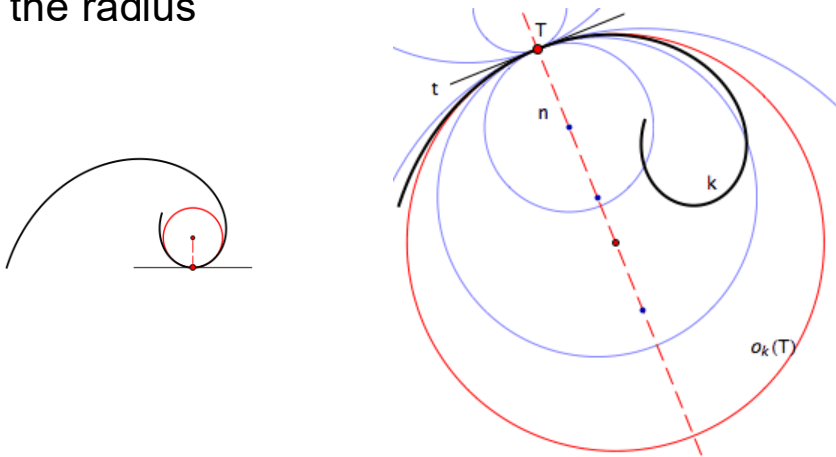
Curvature in 2D



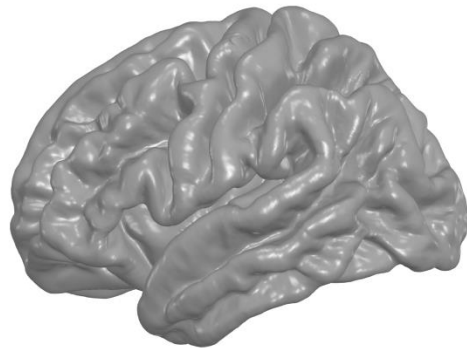
Convex vs concave curves



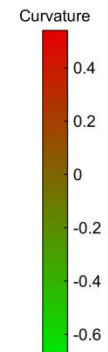
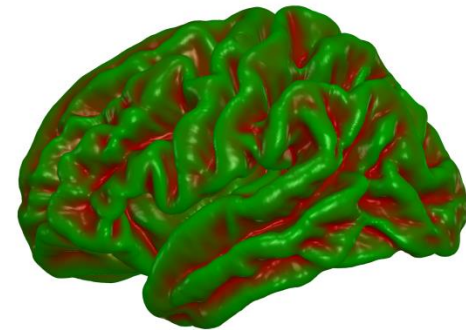
Curvature is usually defined as the inverse of the radius



Curvature maps and sulcal basins



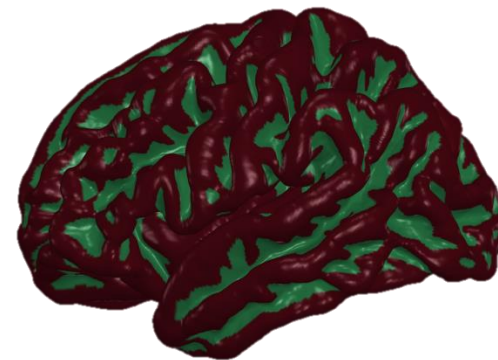
Pial cortical Surface



Curvature Map



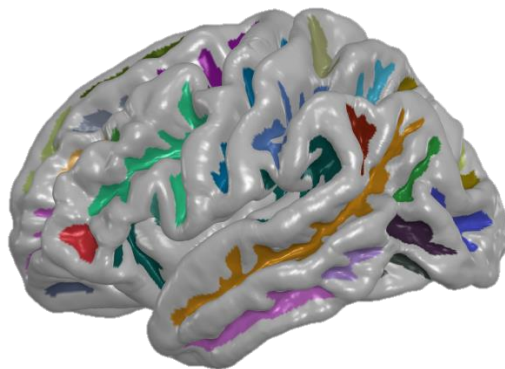
Thresholding



Green Regions: Sulcal Basins

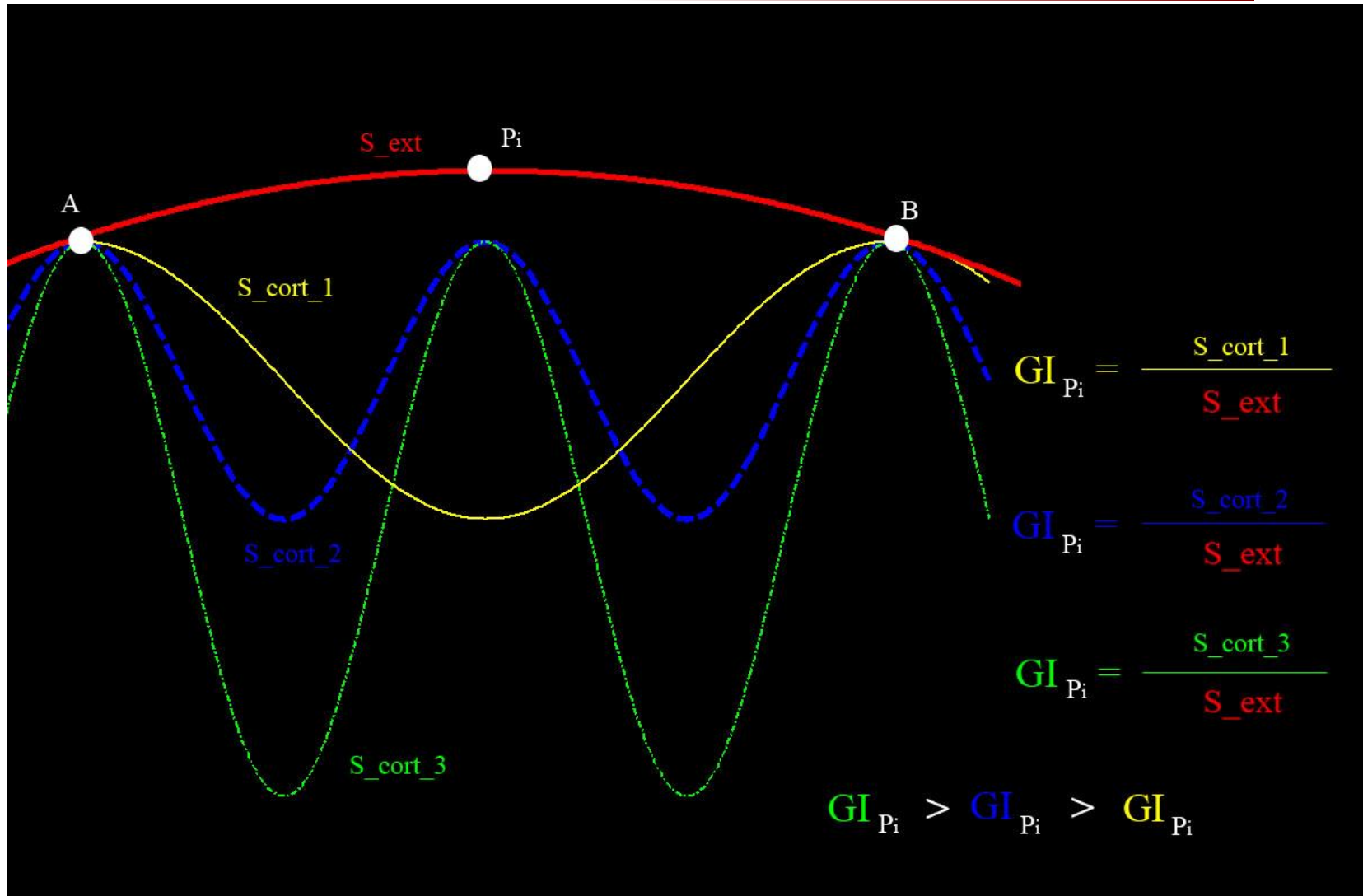


Labelling

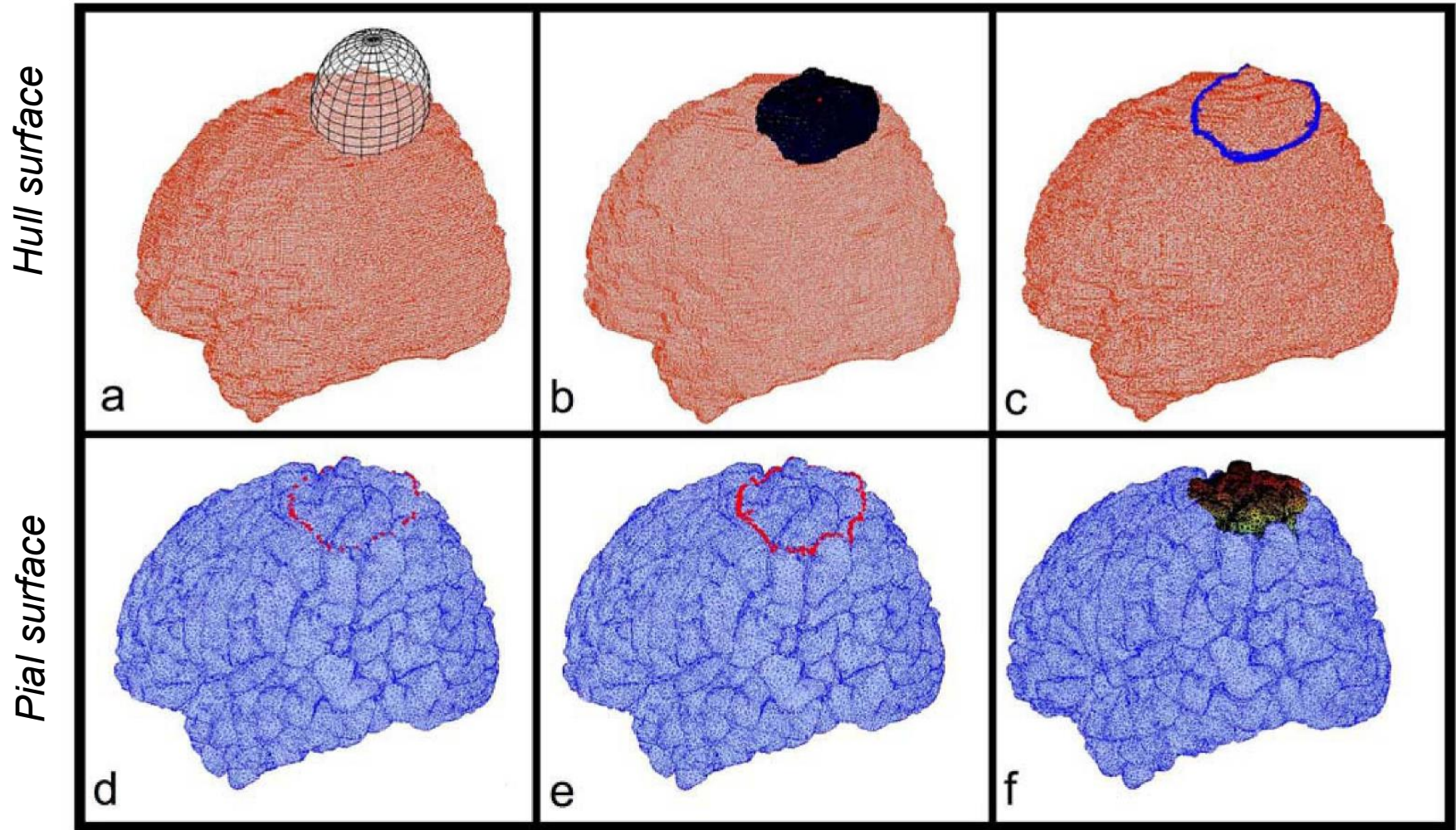


Sulcal Basins (Pial Surface)

Local gyrification index

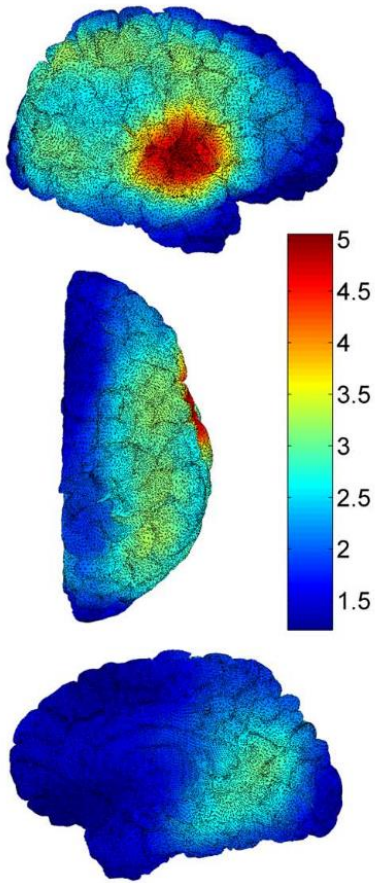


Local gyrification index

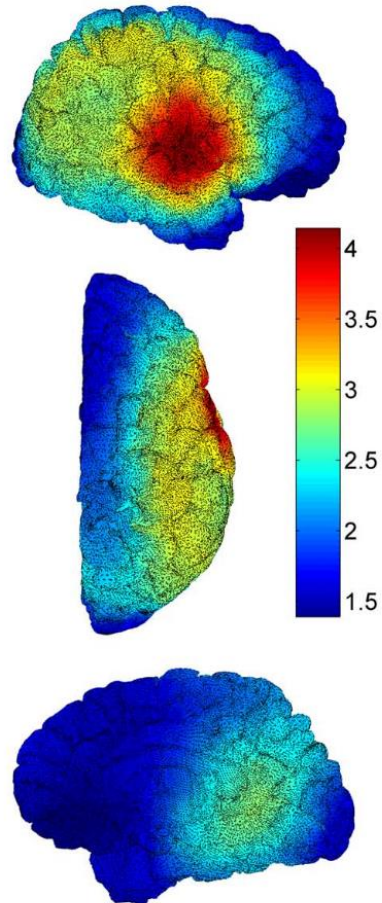


Local gyrification index

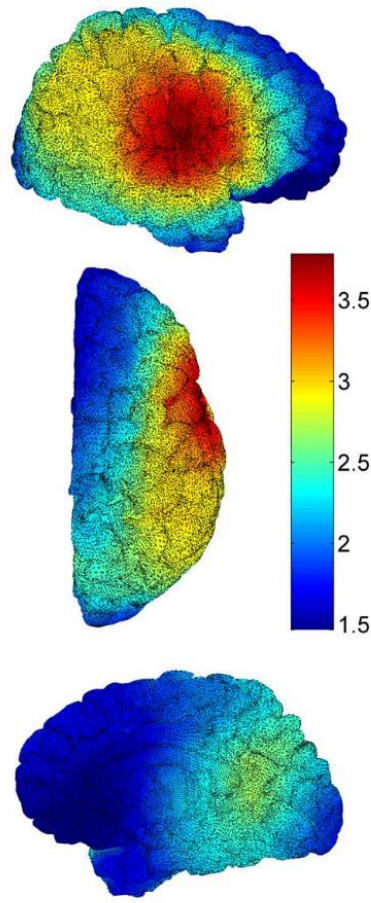
Radius = 20 mm



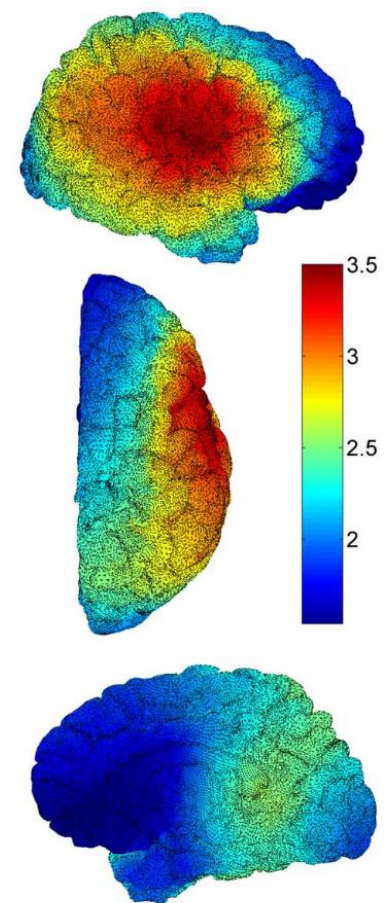
Radius = 25 mm



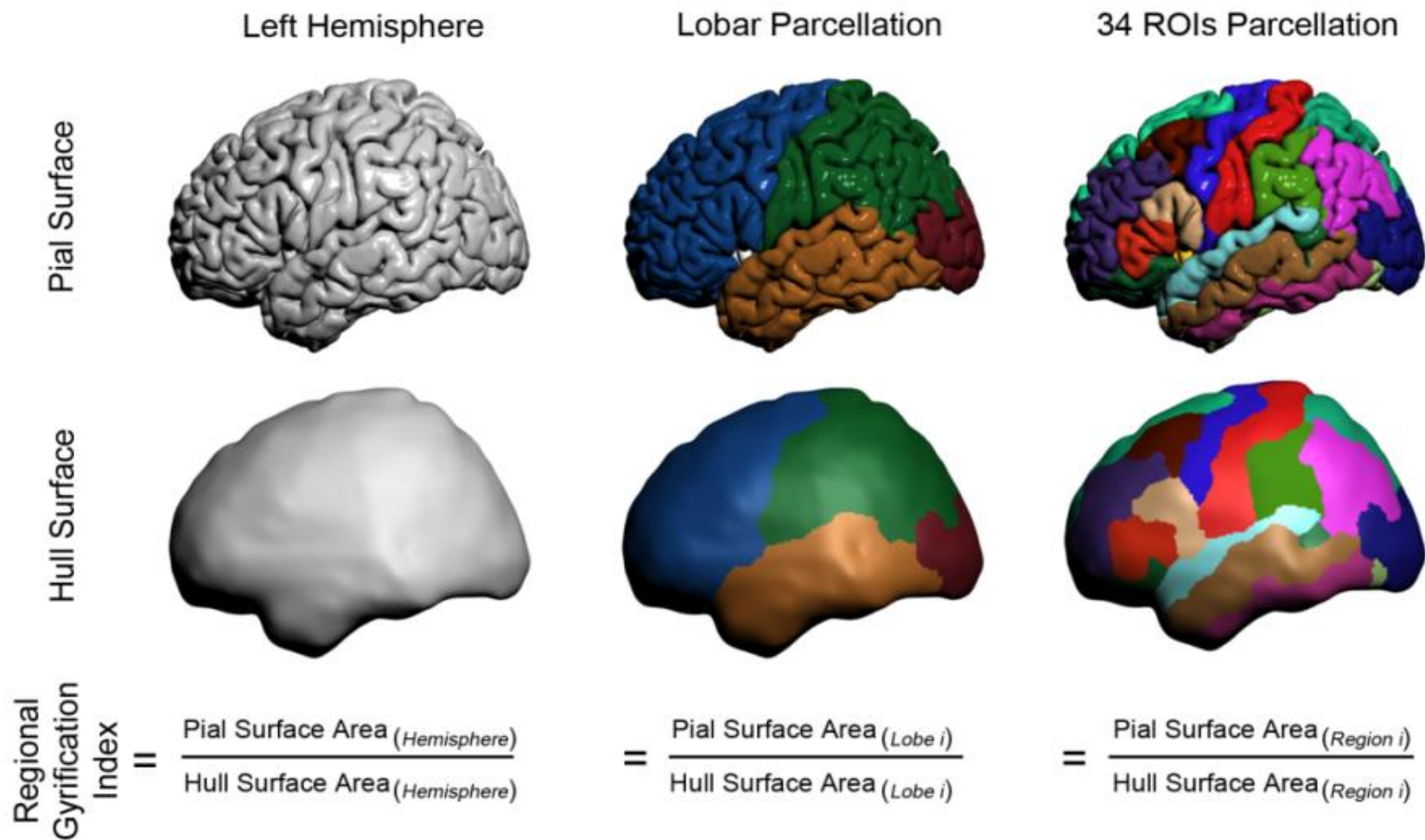
Radius = 30 mm



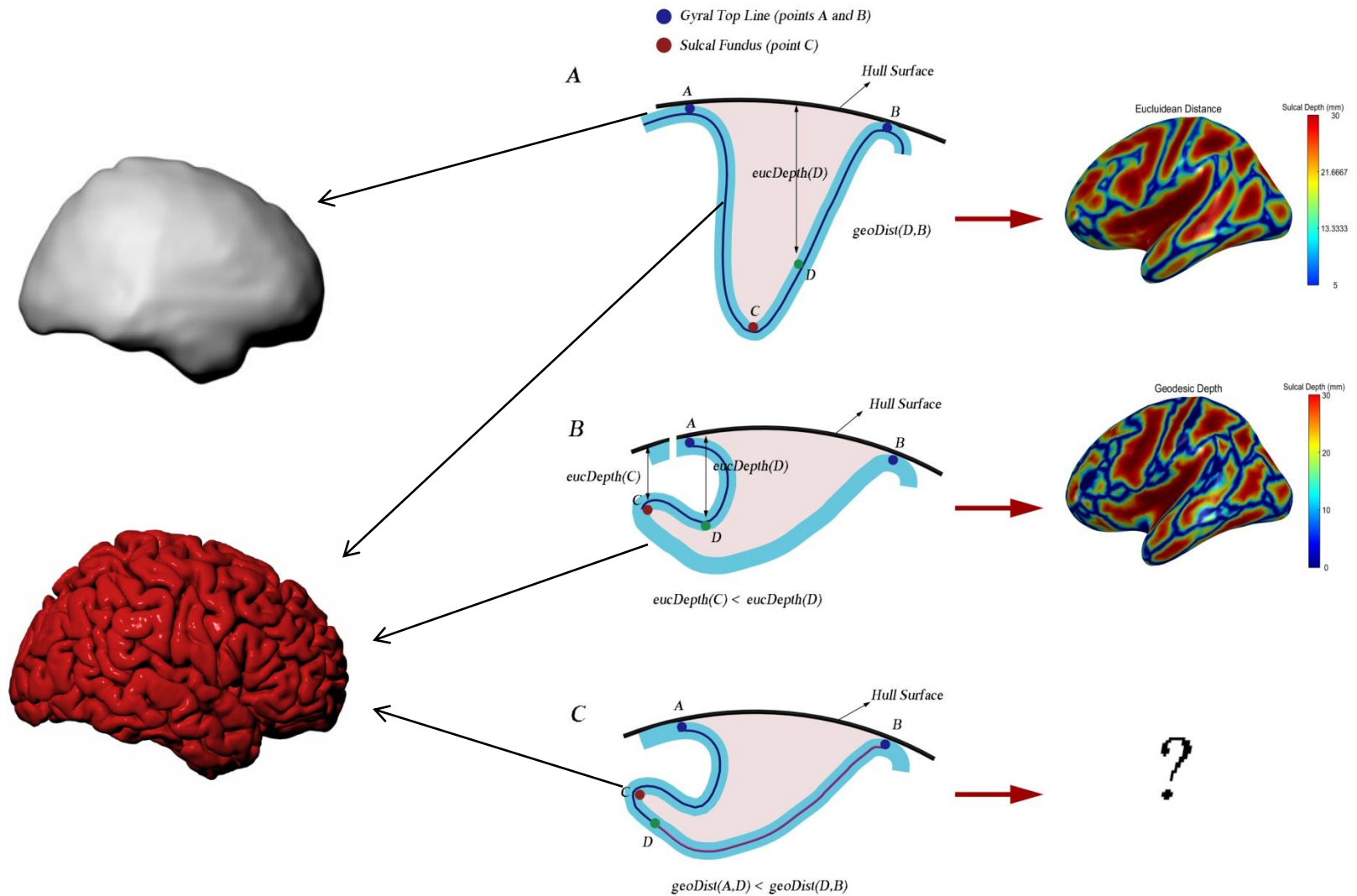
Radius = 35 mm



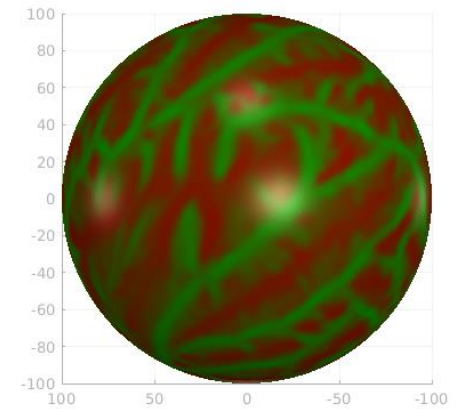
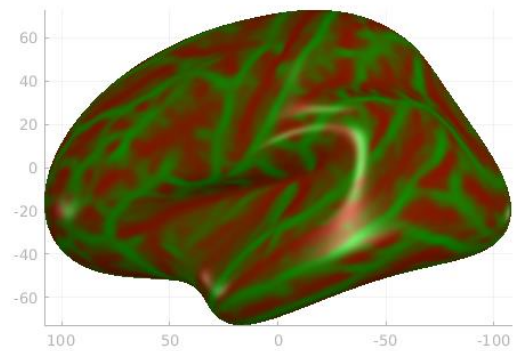
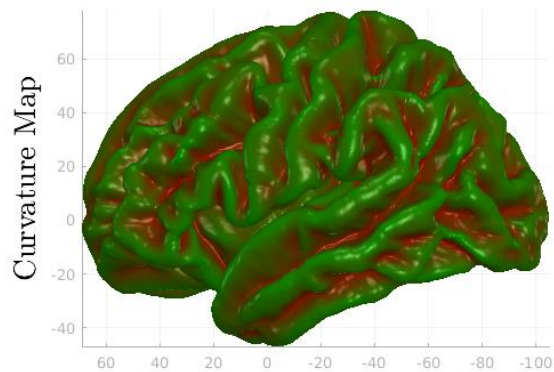
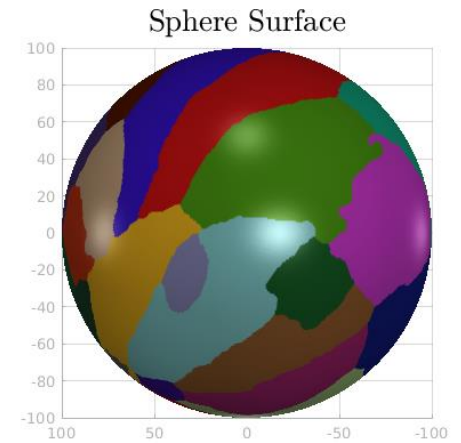
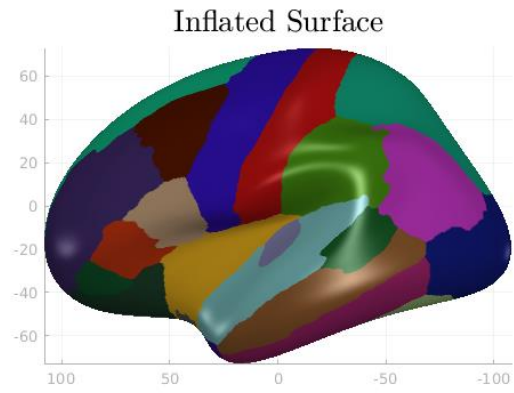
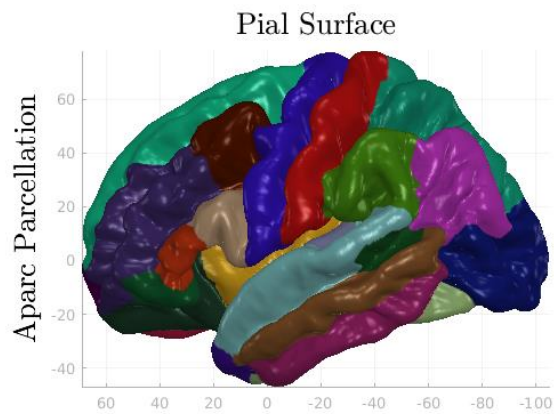
Regional gyrification index



Sulcal depth

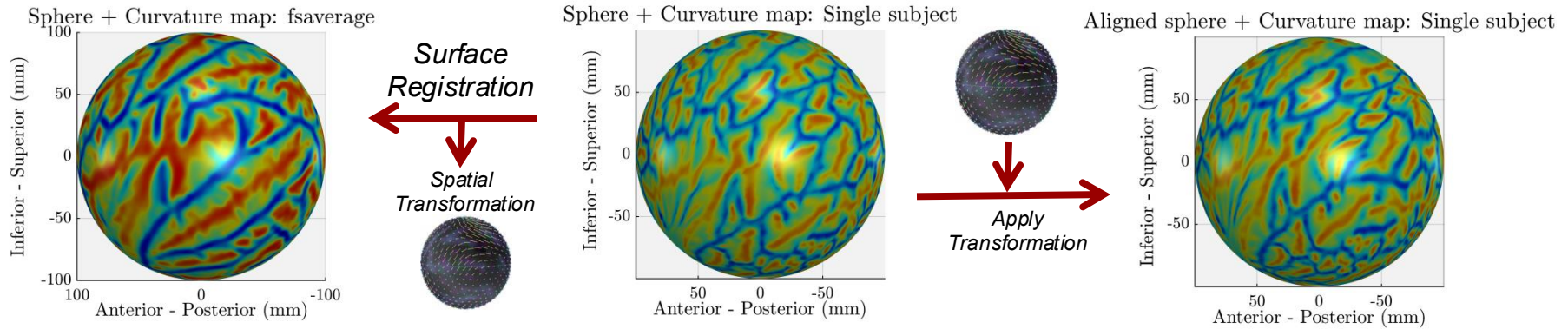


Surface mapping

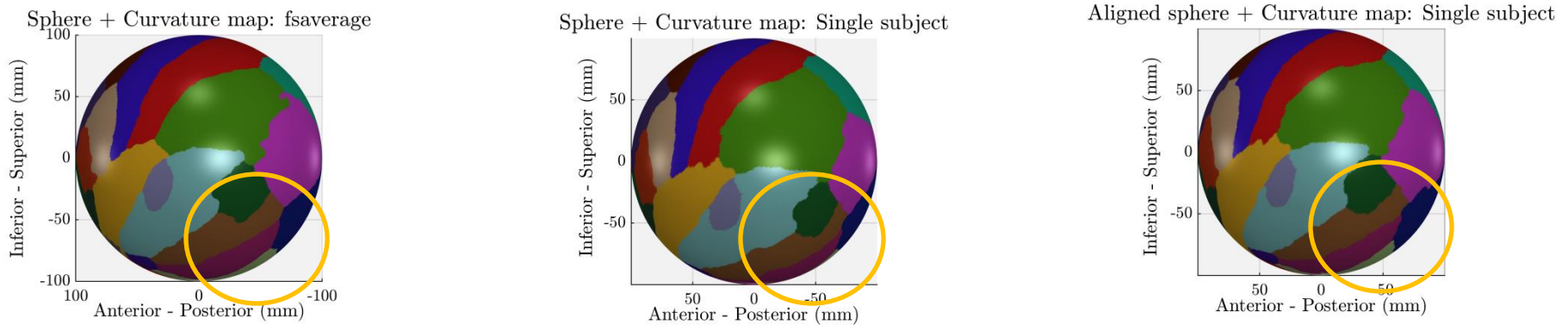


Surface alignment

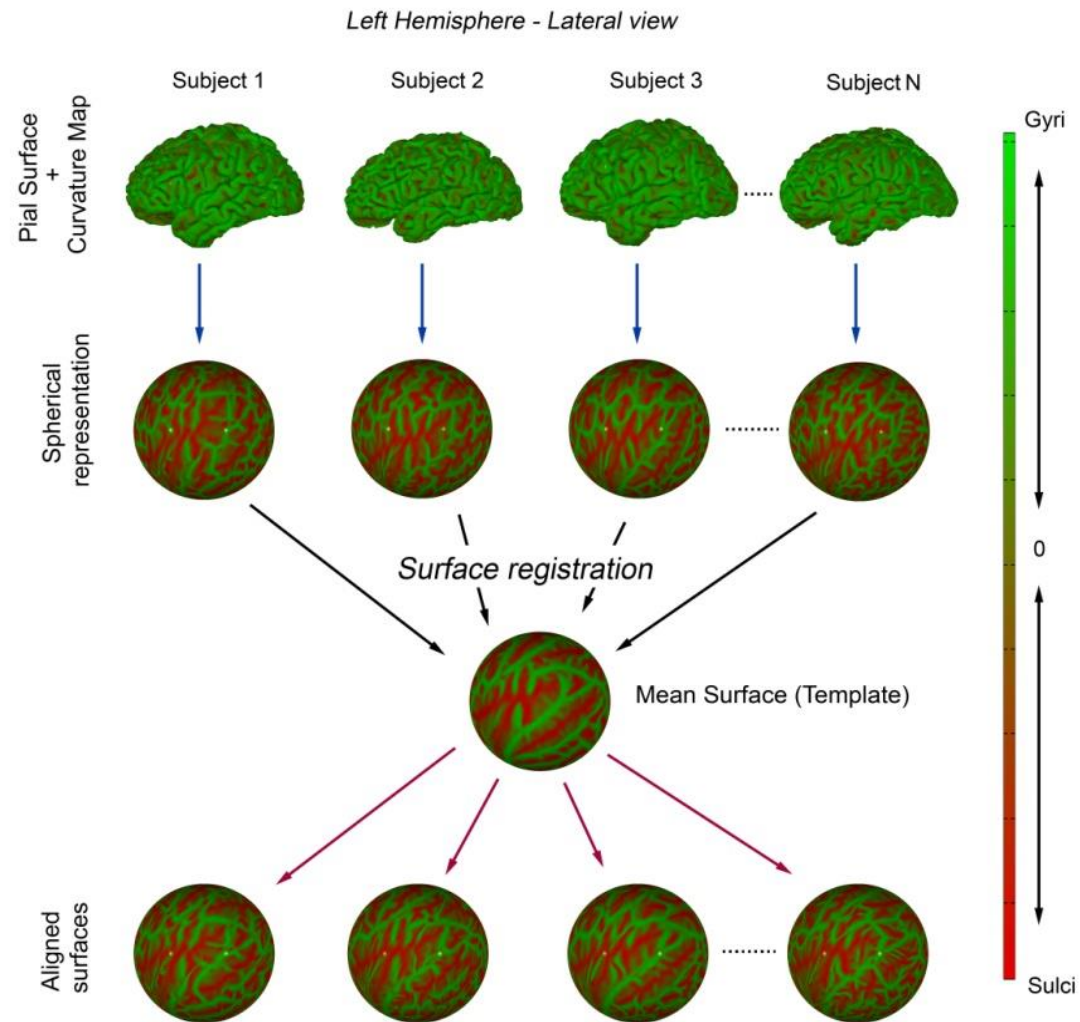
Surface alignment workflow



Cortical parcellation over the sphere



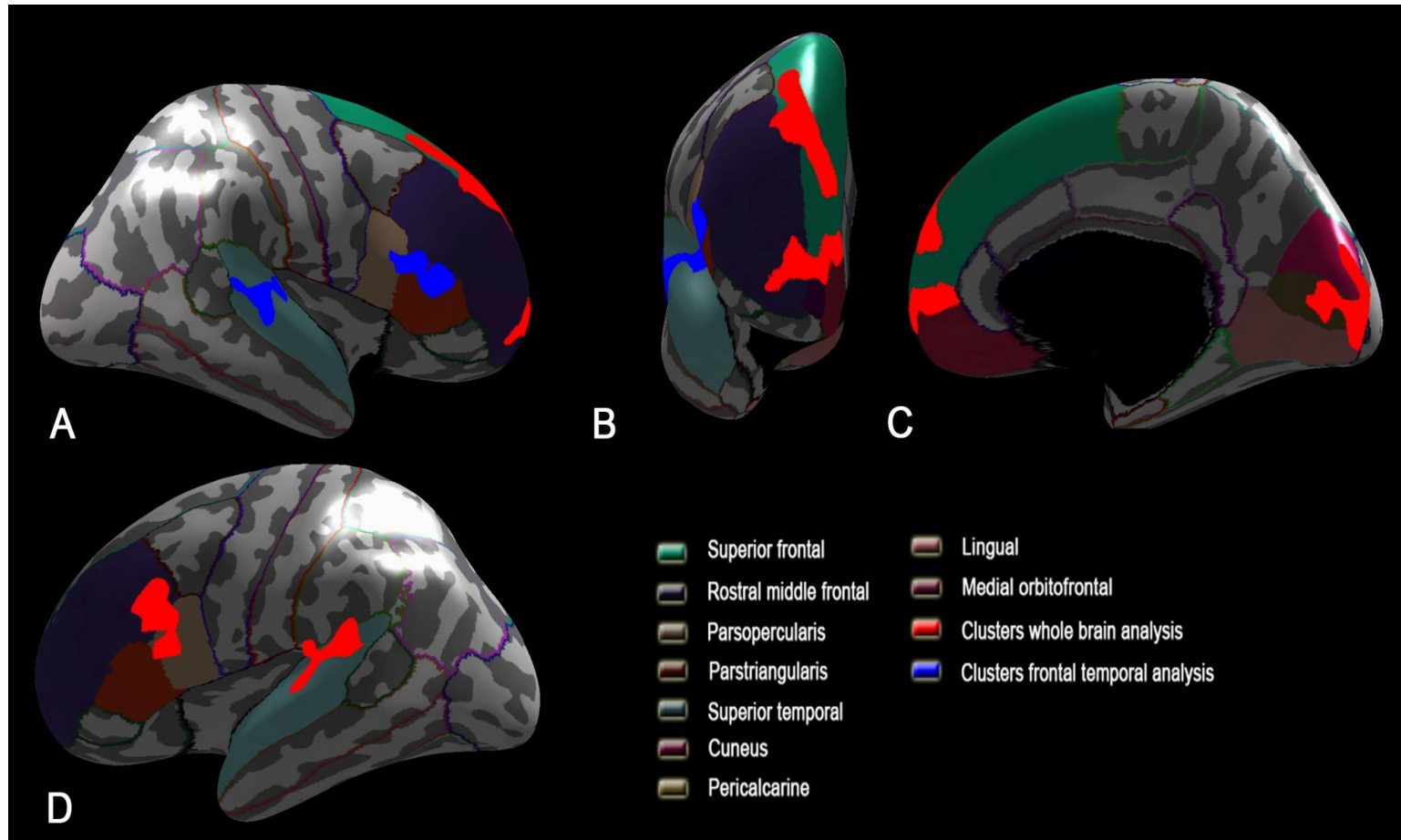
Group alignment



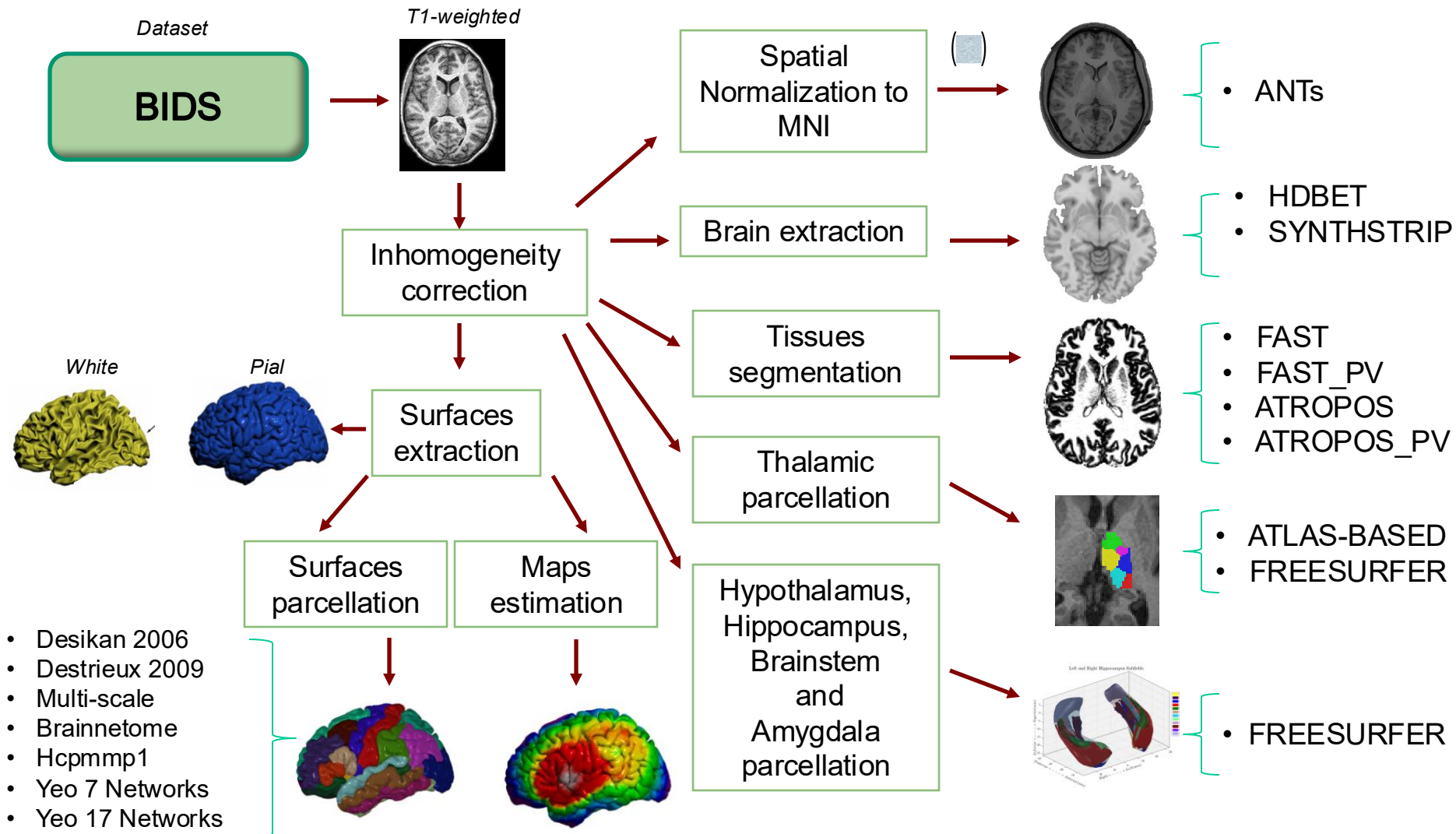
Geodesic smoothing



Surface-Based Morphometry results



Processing for T1-weighted images



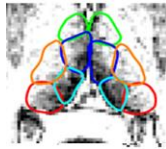
Derivatives: Metrics

Regional metrics

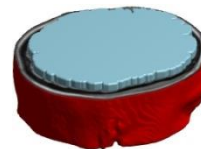
Surface Area ^[1]



Mean GM concentration



Intracranial volume

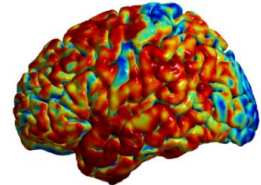


Vertex/voxel-wise metrics

GM Concentration^[5]



Cortical thickness ^[6]



Gyrification Index ^[2,3]

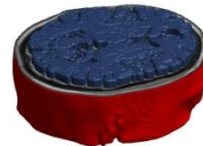
Pial Surface

Hull Surface

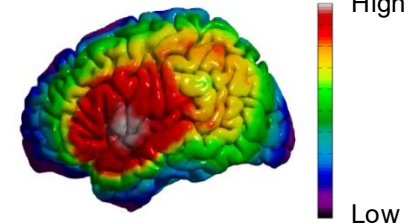


$$GI = \frac{\text{Pial Surface Area}}{\text{Hull Surface Area}} > 1$$

Brain volume

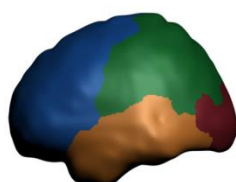
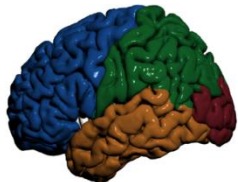


Local Gyrification Index ^[7]



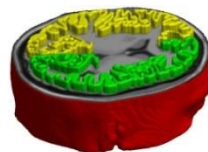
Pial Surface

Hull Surface

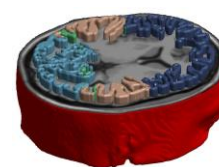


$$rGI = \frac{\text{Pial Surface Area}_{(\text{Lobe})}}{\text{Hull Surface Area}_{(\text{Lobe})}} > 1$$

Tissues



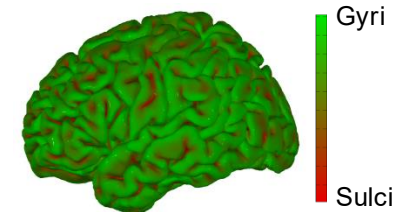
Lobes



Structures



Curvature Index ^[8]



[1] Desikan et al. 2006. Neuroimage

[2] Van Essen et al. 2001. JAMIA

[3] Su et al. 2013. Human Brain Mapping

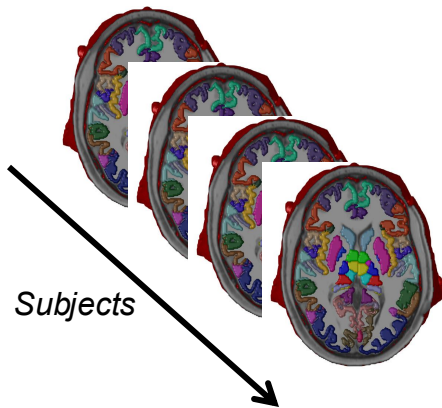
[6] Roche et al. 2014. MICCAI

[6] Fischl et al. 2000. PNAS

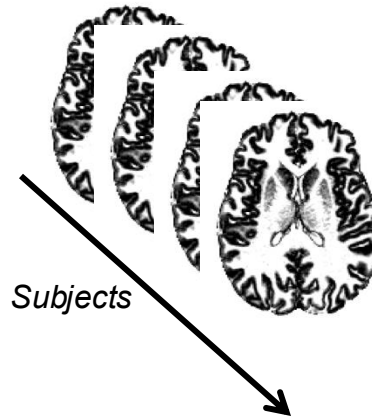
[7] Schaer et al. 2008. IEEE Trans. Med. Imag.

[8] Dale et al. 1999. Neuroimage

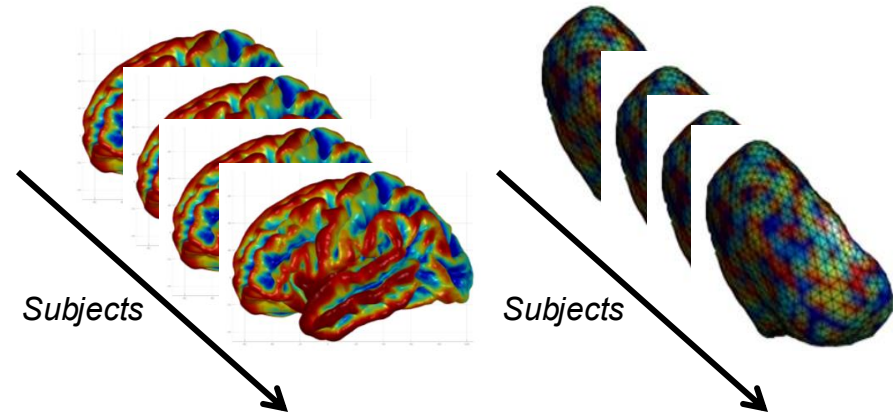
ROI analysis



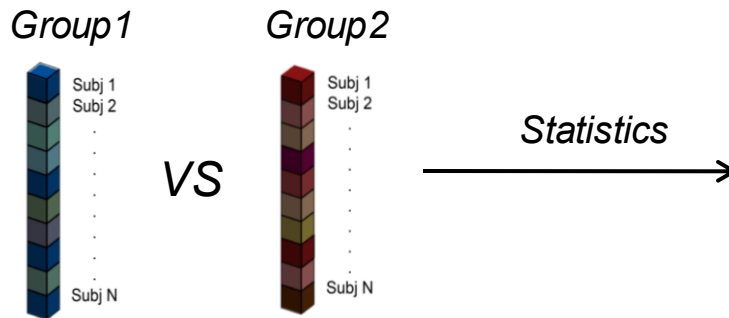
Voxel-Based Morphometry



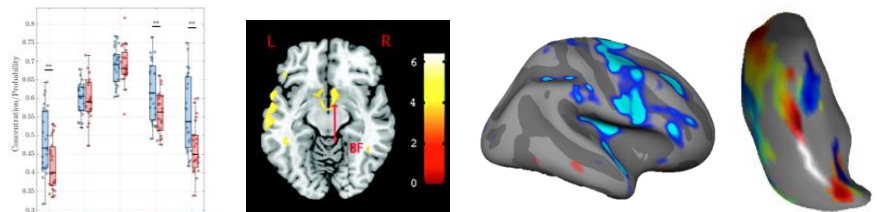
Surface-Based Morphometry



For each ROI, voxel or vertex

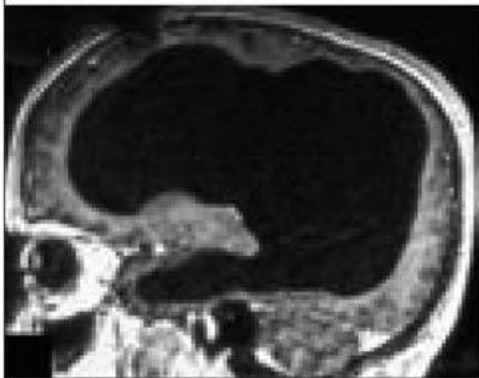
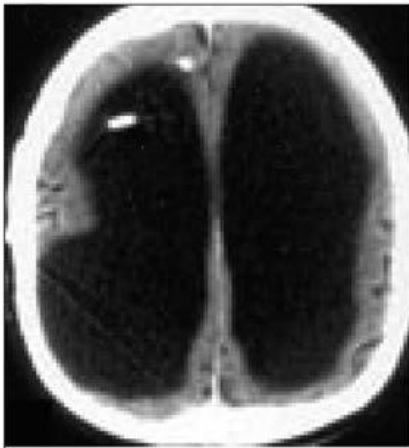


Results

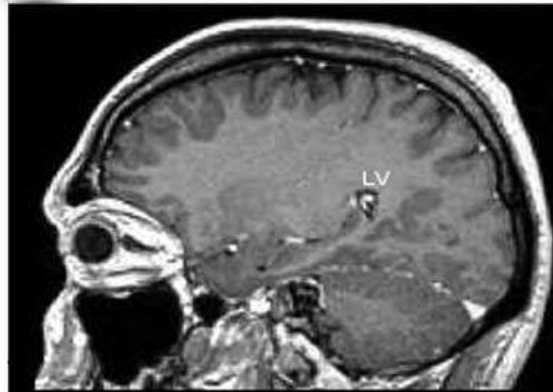
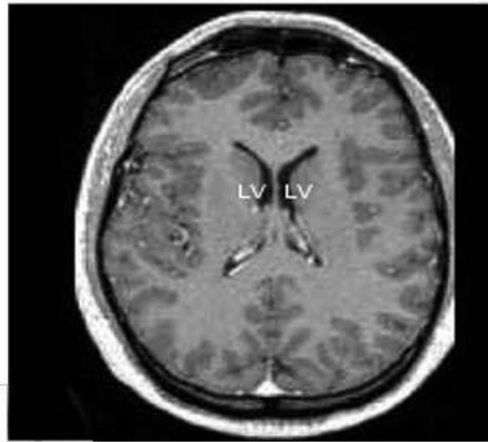


Interesting cases (I)

Sex: Male
Age: 44 yo
IQ: 75 (below the mean)



Healthy control



The Lancet

Interesting cases (II)

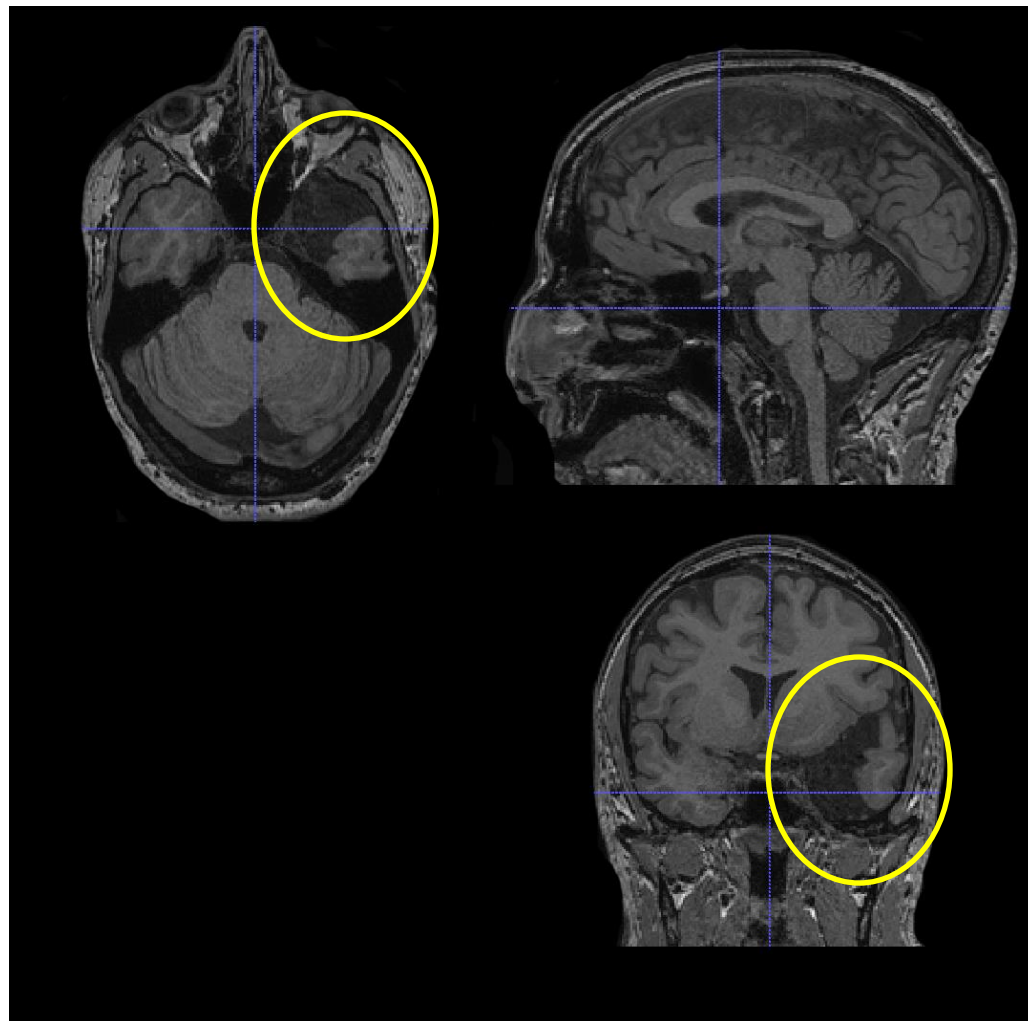
Sex: Female
Age: 24 oy



Healthy control

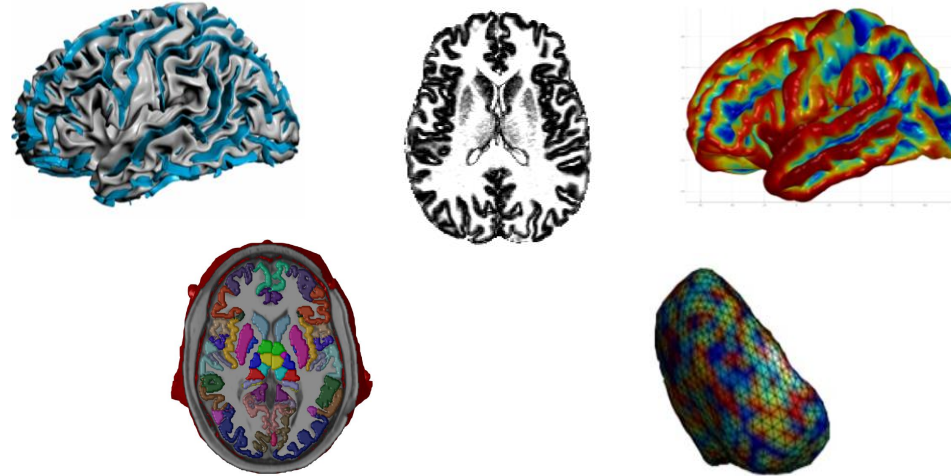
Interesting cases(III) HGGM

Sex: Male
Age: 24 yo



References

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- Mechelli et al. (2005) *Voxel-based morphometry of the human brain...* Current Medical Imaging Reviews 1(2)
- Ashburner & Friston (2009) *Computing average shaped tissue probability templates*. NeuroImage 45(2):333-341
- Dale, A. M., B. Fischl, et al. (1999). "Cortical surface-based analysis. I. Segmentation and surface reconstruction." Neuroimage. 9(2): 179-194.
- Davatzikos, C. and N. Bryan (1996). "Using a deformable surface model to obtain a shape representation of the cortex." IEEE Transactions on Medical Imaging. 15(6): 785-795.
- Fischl, B., M. I. Sereno, et al. (1999). "Cortical surface-based analysis. II: Inflation, flattening, and a surface-based coordinate system." Neuroimage. 9(2): 195-207.
- Thomas E. Nichols: <http://www.sph.umich.edu/~nichols/FDR/>
- Jeanette Mumford: <https://www.youtube.com/channel/UCZ7gF0zm35FwrFpDND6DWeA>
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MRI Processing Theory: Structural Morphometry

Yasser Alemán-Gómez

Radiology department. CHUV, Switzerland
October 1st, 2026